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GAS SENSOR, METHOD FOR DETERMINING ADJUSTMENT VOLTAGE TARGET VALUE FOR CONTROL OF SENSOR ELEMENT, METHOD FOR MANUFACTURING SENSOR ELEMENT, AND METHOD FOR MANUFACTURING GAS SENSOR

Abstract

A gas sensor includes a sensor element and a control device, the sensor element including: a measurement pump cell, an adjustment pump cell including an inner adjustment electrode disposed in an oxygen concentration adjustment chamber, and a reference electrode. The control device is configured to control the adjustment pump cell so that an adjustment voltage that is a voltage across the reference electrode and the inner adjustment electrode reaches an adjustment voltage target value, and let $V1^*$ [mV] be the adjustment voltage target value, then following Expression (1) and Expression (2) are satisfied: $V1^* \leq -25.7 \cdot R + 436.0$ (1) $V1^* \geq -35.1 \cdot R + 406.6$ (2), where $R = Lb \cdot Lb / La$, La is a limiting current value [mA] of an adjustment pump current which flows through the adjustment pump cell and Lb is a limiting current value [μ A] of a measurement pump current which flows through the measurement pump cell.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] This application is a continuation application of PCT/JP2023/031485, filed on Aug. 30, 2023, which claims the benefit of priority of Japanese Patent Application No. JP2022-188981, filed on Nov. 28, 2022, the entire contents of which are incorporated herein by reference.

BACKGROUND OF THE INVENTION

1. Field of the Invention

[0002] The present invention relates a gas sensor, a method for determining an adjustment voltage target value for control of a sensor element, a method for manufacturing a sensor element, and a method for manufacturing a gas sensor.

2. Description of the Related Art

[0003] Hitherto, a gas sensor that detects the concentration of a specific gas, such as NO_x, in a measurement-object gas, such as the exhaust gas of an automobile, is known. For example, PTL 1 describes a gas sensor comprising: an element body which includes an oxygen-ion-conductive solid electrolyte layer, and is internally provided with a measurement-object gas flow portion that introduces a measurement-object gas and causes the measurement-object gas to flow therethrough; and a plurality of electrodes disposed in the element body. When the concentration of NO_x is detected by the gas sensor, pumping-out or pumping-in of oxygen is first performed between an oxygen concentration adjustment chamber of the measurement-object gas flow portion, and the outside of a sensor element, and the oxygen concentration in the oxygen concentration adjustment chamber is adjusted. The measurement-object gas with the oxygen concentration adjusted reaches a measurement chamber provided downstream of the oxygen concentration adjustment chamber of the measurement-object gas flow portion. In the measurement chamber, the NO_x in measurement-object gas is reduced in the periphery of a measurement electrode disposed in the measurement chamber. Then, feedback control is performed on a measurement pump cell so that a voltage V₂ generated across the measurement electrode and a reference electrode reaches a predetermined target value, and the oxygen around the measurement electrode is pumped out. The concentration of NO_x in the measurement-object gas is detected based on pump current I_{p2} which flows then.

CITATION LIST

Patent Literature

[0004] PTL 1: Japanese Unexamined Patent Application Publication No. 2022-091669

SUMMARY OF THE INVENTION

[0005] When a specific gas concentration in a measurement-object gas is measured by a gas sensor, even with the same specific gas concentration, the value of a pump current flowing through a measurement pump cell may vary due to a variation of the water concentration in the measurement-object gas. Thus, the accuracy of measurement of the specific gas concentration may be reduced due to the variation of the water concentration in the measurement-object gas.

[0006] The present invention has been devised to solve such a problem, and it is a main object to prevent reduction in the accuracy of measurement due to the variation of the water concentration in the measurement-object gas.

[0007] The present invention employs the following device to achieve the above-described object.

[0008] [1] A gas sensor of the present invention is a gas sensor comprising a sensor element, and a control device, the sensor element including: an element body that includes an oxygen-ion-conductive solid electrolyte layer, and is internally provided with a measurement-object gas flow portion that introduces a measurement-object gas and causes the measurement-object gas to flow therethrough; a measurement pump cell including an inner measurement electrode disposed in a measurement chamber of the measurement-object gas flow portion, the measurement pump cell being configured to pump oxygen from the measurement chamber to an outside of the element body; an adjustment pump cell including an inner adjustment electrode disposed in an oxygen concentration adjustment chamber located upstream of the measurement chamber in the measurement-object gas flow portion, the adjustment pump cell being configured to adjust an oxygen concentration in the oxygen concentration adjustment chamber; and a reference electrode disposed inside the element body so as to be in contact with a reference gas which serves as a reference for detection of a specific gas concentration that is a concentration of a specific gas in the measurement-object gas. The control device is configured to perform an adjustment-pump control process and a measurement-pump control process, the adjustment-pump control process for adjusting an oxygen concentration in the oxygen concentration adjustment chamber by controlling the adjustment pump cell so that an adjustment voltage that is a voltage across the reference electrode and the inner adjustment electrode reaches an adjustment voltage target value, the measurement-pump control process for pumping out oxygen from the measurement chamber by controlling the measurement pump cell so that a measurement voltage that is a voltage across the reference electrode and the inner measurement electrode reaches a measurement voltage target value, and let $V1^*$ [mV] be the adjustment voltage target value, then following Expression (1) and Expression (2) are satisfied:

[00001] $V1^* \leq -25.7 * R + 436.$ (1) $V1^* \geq -35.1 * R + 406.6$ (2) [0009] where

$R = Lb * Lb / La$, [0010] La is a limiting current value [mA] of an adjustment pump current which flows through the adjustment pump cell when the adjustment pump cell pumps out oxygen from the oxygen concentration adjustment chamber with the sensor element exposed to the measurement-object gas which is a nitrogen-based gas with an oxygen concentration of 20.5%, a water concentration of 3%, and an NO concentration of 0 ppm, and Lb is a limiting current value [μ A] of a measurement pump current which flows through the measurement pump cell when the measurement pump cell pumps out oxygen from the measurement chamber with the sensor element exposed to the measurement-object gas which is a nitrogen-based gas with an oxygen concentration of 0%, a water concentration of 3%, and an NO concentration of 500 ppm.

[0011] In this gas sensor, the control device performs the adjustment-pump control process and the measurement-pump control process. The oxygen concentration in the oxygen concentration adjustment chamber is adjusted by performing the adjustment-pump control process, then the measurement-object gas reaches the measurement chamber. Then the measurement-pump control process is performed, thereby causing the measurement pump current to flow to pump out oxygen from the measurement chamber. The specific gas concentration in the measurement-object gas can be measured based on the measurement pump current. In the gas sensor, let $V1^*$ [mV] be the adjustment voltage target value in the adjustment-pump control process, then the value of $V1^*$ satisfies the Expression (1) and Expression (2). Consequently, even if the water concentration in the measurement-object gas varies, it is unlikely that the measurement pump current varies due to the water concentration. Therefore, it is possible to prevent reduction in the accuracy of measurement of a specific gas concentration due to a variation of the water concentration in the measurement-

object gas. The inventors have verified this fact by experiments.

[0012] [2] In the above-described gas sensor (the gas sensor according to [1]), Expression (3) and Expression (4) below may be satisfied. In this setting, it is possible to further prevent reduction in the accuracy of measurement of a specific gas concentration due to a variation of the water concentration in the measurement-object gas.

$$[00002] \quad V1 \star \leq -18.9 \star R + 419.2 \quad (3) \quad V1 \star \geq -16.2 \star R + 403.4 \quad (4)$$

[0013] [3] In the above-described gas sensor (the gas sensor according to [1] or [2]), the oxygen concentration adjustment chamber may include a first internal cavity, and a second internal cavity provided downstream of the first internal cavity and upstream of the measurement chamber, the adjustment pump cell may include a main pump cell configured to adjust an oxygen concentration in the first internal cavity, and an auxiliary pump cell configured to adjust an oxygen concentration in the second internal cavity, the inner adjustment electrode may include an inner main pump electrode that is disposed in the first internal cavity and constitutes part of the main pump cell, and an inner auxiliary pump electrode that is disposed in the second internal cavity and constitutes part of the auxiliary pump cell, the adjustment voltage may be a voltage across the reference electrode and the inner auxiliary pump electrode, and the limiting current value I_a may be a limiting current value of a main pump current which flows through the main pump cell when the main pump cell pumps out oxygen from the first internal cavity with the sensor element exposed to the measurement-object gas which is a nitrogen-based gas with an oxygen concentration of 20.5%, a water concentration of 3%, and an NO concentration of 0 ppm.

[0014] [4] A method for determining an adjustment voltage target value for control of a first sensor element of the present invention, the sensor element including: an element body that includes an oxygen-ion-conductive solid electrolyte layer, and is internally provided with a measurement-object gas flow portion that introduces a measurement-object gas and causes the measurement-object gas to flow therethrough; a measurement pump cell including an inner measurement electrode disposed in a measurement chamber of the measurement-object gas flow portion, the measurement pump cell being configured to pump oxygen from the measurement chamber to an outside of the element body; an adjustment pump cell including an inner adjustment electrode disposed in an oxygen concentration adjustment chamber located upstream of the measurement chamber in the measurement-object gas flow portion, the adjustment pump cell being configured to adjust an oxygen concentration in the oxygen concentration adjustment chamber; and a reference electrode disposed inside the element body so as to be in contact with a reference gas which serves as a reference for detection of a specific gas concentration that is a concentration of a specific gas in the measurement-object gas. The adjustment voltage target value is a target value of an adjustment voltage when a specific gas concentration that is a concentration of a specific gas in the measurement-object gas is detected using the sensor element, the adjustment voltage being a voltage across the reference electrode and the inner adjustment electrode at a time of feedback control of the adjustment pump cell to adjust the oxygen concentration in the oxygen concentration adjustment chamber, the method including a step (a) of determining the adjustment voltage target value so that lower a diffusion resistance B relative to a diffusion resistance A, the adjustment voltage target value tends to be lower, where the diffusion resistance A is a diffusion resistance from an outside of the sensor element to the oxygen concentration adjustment chamber, and the diffusion resistance B is a diffusion resistance from the outside of the sensor element to the oxygen measurement chamber.

[0015] In the method for determining an adjustment voltage target value for control of a sensor element, the adjustment voltage target value is determined so that smaller the diffusion resistance B relative to the diffusion resistance A, the adjustment voltage target value tends to be lower, where the diffusion resistance A is the diffusion resistance from the outside of the sensor element to the oxygen concentration adjustment chamber, and the diffusion resistance B is the diffusion resistance

from the outside of the sensor element to the measurement chamber. The inventors have found that the gas sensor has an optimal point for the adjustment voltage target value, the optimal point being capable of preventing reduction in the accuracy of measurement of a specific gas concentration due to a variation of the water concentration in the measurement-object gas. In addition, the inventors have found that such an optimal point for the adjustment voltage target value changes by the balance between the diffusion resistance A and the diffusion resistance B. More specifically, the inventors have found that the smaller the diffusion resistance B relative to the diffusion resistance A in the sensor element, the optimal point for the adjustment voltage target value tends to be lower. Thus, the adjustment voltage target value is easily determined to an optimal point or a value close to an optimal point by determining the adjustment voltage target value so that the smaller the diffusion resistance B relative to the diffusion resistance A, the adjustment voltage target value tends to be lower. Controlling the sensor element based on thus determined adjustment voltage target value facilitates prevention of reduction in the accuracy of measurement of a specific gas concentration due to a variation of the water concentration in the measurement-object gas.

[0016] [5] In the above-described method for determining an adjustment voltage target value (the method for determining an adjustment voltage target value according to [4]), the step (a) may include: a step (a1) of measuring, as a value correlated with the diffusion resistance A, a limiting current value La of an adjustment pump current which flows through the adjustment pump cell when the adjustment pump cell pumps out oxygen from the oxygen concentration adjustment chamber, and as a value correlated with the diffusion resistance B, a limiting current value Lb of a measurement pump current which flows through the measurement pump cell when the measurement pump cell pumps out oxygen from the measurement chamber; and a step (a2) of determining the adjustment voltage target value so that for the limiting current values La, Lb measured in the step (a1), greater the limiting current value Lb relative to the limiting current value La, the adjustment voltage target value tends to be lower. In this manner, instead of measuring the diffusion resistance A and the diffusion resistance B, the limiting current value La and the limiting current value Lb are measured, and the adjustment voltage target value can be determined to an optimal point or a value close to an optimal point based on the measured values.

[0017] [6] In the above-described method for determining an adjustment voltage target value (the method for determining an adjustment voltage target value according to [5]), in the step (a2), the adjustment voltage target value may be determined so that the adjustment voltage target value tends to be lower for a greater value R expressed below. The inventors have found that the value R measured using the sensor element has a linear relationship with the optimal point of the adjustment voltage target value. Thus, the adjustment voltage target value is easily determined to an optimal point or a value close to an optimal point by determining the adjustment voltage target value using the value R so that the adjustment voltage target value tends to be lower for a larger value R.

[00003] $R = Lb \star Lb / La$.

[0018] [7] In the above-described method for determining an adjustment voltage target value (the method for determining an adjustment voltage target value according to [5] or [6]), in the step (a2), the adjustment voltage target value may be determined as a value corresponding to the limiting current values La, Lb measured in the step (a1) based on a prepared corresponding relationship in advance between the limiting current values La, Lb and the adjustment voltage target value. In this manner, an appropriate corresponding relationship is prepared in advance, for example, by conducting experiments on multiple sensor elements, thus in the step (a2), an appropriate adjustment voltage target value can be easily determined using the corresponding relationship.

[0019] [8] In the above-described method for determining an adjustment voltage target value (the method for determining an adjustment voltage target value according to any one of [5] to [7]), the oxygen concentration adjustment chamber may include a first internal cavity, and a second internal cavity provided downstream of the first internal cavity and upstream of the measurement chamber,

the adjustment pump cell may include a main pump cell configured to adjust an oxygen concentration in the first internal cavity, and an auxiliary pump cell configured to adjust an oxygen concentration in the second internal cavity, the inner adjustment electrode may include an inner main pump electrode that is disposed in the first internal cavity and constitutes part of the main pump cell, and an inner auxiliary pump electrode that is disposed in the second internal cavity and constitutes part of the auxiliary pump cell, the adjustment voltage may be a voltage across the reference electrode and the inner auxiliary pump electrode, and the limiting current value I_a may be a limiting current value of a main pump current which flows through the main pump cell when the main pump cell pumps out oxygen from the first internal cavity.

[0020] [9] A method for determining an adjustment voltage target value for control of a second sensor element of the present invention, the sensor element including: an element body that includes an oxygen-ion-conductive solid electrolyte layer, and is internally provided with a measurement-object gas flow portion that introduces a measurement-object gas and causes the measurement-object gas to flow therethrough; a measurement pump cell including an inner measurement electrode disposed in a measurement chamber of the measurement-object gas flow portion, the measurement pump cell being configured to pump oxygen from the measurement chamber to an outside of the element body; an adjustment pump cell including an inner adjustment electrode disposed in an oxygen concentration adjustment chamber located upstream of the measurement chamber in the measurement-object gas flow portion, the adjustment pump cell being configured to adjust an oxygen concentration in the oxygen concentration adjustment chamber; and a reference electrode disposed inside the element body so as to be in contact with a reference gas which serves as a reference for detection of a specific gas concentration that is a concentration of a specific gas in the measurement-object gas. The sensor element is configured to, when a specific gas concentration that is a concentration of a specific gas in the measurement-object gas is detected, perform an adjustment-pump control process and a measurement-pump control process, the adjustment-pump control process for adjusting the oxygen concentration in the oxygen concentration adjustment chamber by controlling the adjustment pump cell so that an adjustment voltage that is a voltage across the reference electrode and the inner adjustment electrode reaches the adjustment voltage target value, the measurement-pump control process for pumping out oxygen from the measurement chamber by controlling the measurement pump cell so that a measurement voltage that is a voltage across the reference electrode and the inner measurement electrode reaches a measurement voltage target value, the method including: a step (b1) of setting the measurement voltage target value to a predetermined value, setting the adjustment voltage target value to a temporary target value, and measuring a first measurement pump current and a second measurement pump current, the first measurement pump current being caused to flow through the measurement pump cell when the adjustment-pump control process and the measurement-pump control process are performed on a first gas as the measurement-object gas, the first gas having a water concentration of a first concentration Ch_1 , the second measurement pump current being caused to flow through the measurement pump cell when the adjustment-pump control process and the measurement-pump control process are performed on a second gas as the measurement-object gas, the second gas having a water concentration of a second concentration Ch_2 lower than the first concentration Ch_1 ; and a step (b2) of changing the temporary target value and performing the step (b1) for multiple times, and determining the temporary target value to be the adjustment voltage target value for control of the sensor element, the temporary target value causing a discrepancy between the first measurement pump current and the second measurement pump current to fall within a predetermined minute range.

[0021] In the method for determining an adjustment voltage target value for control of a sensor element, in the step (b1), the adjustment voltage target value is set to a temporary target value, and the first measurement pump current and the second measurement pump current are measured, which flow through when the adjustment-pump control process and the measurement-pump control

process are performed using the first gas and the second gas having different water concentrations. As described above, the gas sensor element has an optimal point for the adjustment voltage target value, the optimal point being capable of preventing reduction in the accuracy of measurement of a specific gas concentration due to a variation of the water concentration in the measurement-object gas. Therefore, when the temporary target value is an optimal point for the adjustment voltage target value or a value close to an optimal point, the first measurement pump current and the second measurement pump current measured in the step (b1) have the same value or values close to each other. Thus, an appropriate adjustment voltage target value can be found by changing the temporary target value, and performing the step (b1) for multiple times. In the step (b2), change in the temporary target value and the step (b1) are performed for multiple times in this manner, and a temporary target value is determined to be the adjustment voltage target value for control of the sensor element, the temporary target value causing the discrepancy between the first measurement pump current and the second measurement pump current to fall within a predetermined minute range. Thus, the adjustment voltage target value can be determined to an optimal point or a value close to an optimal point. Controlling the sensor element based on thus determined adjustment voltage target value can prevent reduction in the accuracy of measurement of a specific gas concentration due to a variation of the water concentration in the measurement-object gas.

[0022] [10] A method for manufacturing a sensor element, including the method (the method for determining an adjustment voltage target value according to any one of [4] to [9]) for determining an adjustment voltage target value for control of the above-described sensor element, and being associated with the adjustment voltage target value, the method comprising: a manufacturing process for manufacturing the sensor element; a target value determination process for determining the adjustment voltage target value by performing, on the manufactured sensor element, the steps of the method for determining the adjustment voltage target value for control of the sensor element; and an association process for associating the manufactured sensor element with the determined adjustment voltage target value.

[0023] As with the above-described method for determining an adjustment voltage target value, with the method for manufacturing a sensor element, it is possible to determine an adjustment voltage target value which can prevent reduction in the accuracy of measurement of a specific gas concentration due to a variation of the water concentration in the measurement-object gas. Since the determined adjustment voltage target value is associated with the sensor element, when a control device is subsequently connected to the sensor element, for example, an adjustment voltage target value suitable for the sensor element can be set.

[0024] [11] The above-described method for manufacturing a sensor includes: the processes of the above-described method for manufacturing a sensor element (the method for manufacturing a sensor element according to [10]); and a connection process for connecting the sensor element associated with the adjustment voltage target value to a control device that feedback-controls the adjustment pump cell so that a voltage across the reference electrode and the inner adjustment electrode reaches the associated adjustment voltage target value.

[0025] With the method for manufacturing a gas sensor, a gas sensor including a control device and a sensor element can be manufactured. In the manufactured gas sensor, the control device controls the adjustment pump cell so that the voltage across the reference electrode and the inner adjustment electrode reaches the adjustment voltage target value determined by the above-described method for determining an adjustment voltage target value, thus it is possible to prevent reduction in the accuracy of measurement of a specific gas concentration due to a variation of the water concentration in the measurement-object gas.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0026] FIG. **1** is a schematic cross-sectional view illustrating an example of a configuration of a gas sensor **100**.

[0027] FIG. **2** is a block diagram illustrating an electrical connection relationship between a control device **95** and cells.

[0028] FIG. **3** is a graph illustrating a relationship between target value $V1^*$, water concentration and pump current I_{p2} .

[0029] FIG. **4** is a graph illustrating a relationship between target value $V1^*$ and $\Delta I_{p2offset}$.

[0030] FIG. **5** is a graph illustrating a relationship between value R , target value $V1^*$ and $\Delta I_{p2offset}$.

[0031] FIG. **6** is a schematic cross-sectional view of a sensor element **201**.

DETAILED DESCRIPTION OF THE INVENTION

First Embodiment

[0032] Next, an embodiment of a gas sensor of the present invention will be described using the drawings. FIG. **1** is a schematic cross-sectional view illustrating an example of a configuration of a gas sensor **100** which is an embodiment of the present invention. FIG. **2** is a block diagram illustrating an electrical connection relationship between a control device **95** and cells. The gas sensor **100** is mounted on, for example, a pipe arranged, such as an exhaust gas pipe of an internal combustion engine. The gas sensor **100** detects the concentration of a specific gas such as NO_x and ammonia in a measurement-object gas which is the exhaust gas from an internal combustion engine. In the present embodiment, the gas sensor **100** measures the NO_x concentration as the specific gas concentration. The gas sensor **100** includes: a sensor element **101** having an elongate rectangular parallelepiped shape; cells **21**, **41**, **50**, **80** to **83** included in the sensor element **101**; and the control device **95** having variable power sources **24**, **46**, **52** and configured to control the entire gas sensor **100**.

[0033] The sensor element **101** is an element having a layered body in which six layers, that is, a first substrate layer **1**, a second substrate layer **2**, a third substrate layer **3**, a first solid electrolyte layer **4**, a spacer layer **5**, and a second solid electrolyte layer **6**, each made up of an oxygen-ion-conductive solid electrolyte layer made of zirconia ($\text{ZrO}_{2.2}$) or the like, are laminated in this order from a lower side in the drawing. The solid electrolyte forming these six layers is a dense, airtight one. The sensor element **101** is manufactured by, for example, applying predetermined processing, printing of a circuit pattern, and the like on a ceramic green sheet corresponding to each layer, then laminating those sheets, and further firing the sheets to be integrated.

[0034] At a tip end portion side of the sensor element **101** (left end portion side in FIG. **1**), a gas inlet port **10**, a first diffusion control section **11**, a buffer space **12**, a second diffusion control section **13**, a first internal cavity **20**, a third diffusion control section **30**, a second internal cavity **40**, a fourth diffusion control section **60**, and a third internal cavity **61** are formed adjacent to each other so as to communicate with each other in this order between the under surface of the second solid electrolyte layer **6** and the top surface of the first solid electrolyte layer **4**.

[0035] The gas inlet port **10**, the buffer space **12**, the first internal cavity **20**, the second internal cavity **40**, and the third internal cavity **61** are spaces of which top parts, bottom parts, and side parts, provided by hollowing the spacer layer **5**, are respectively defined by the under surface of the second solid electrolyte layer **6**, the top surface of the first solid electrolyte layer **4**, and the side surface of the spacer layer **5** inside the sensor element **101**.

[0036] Each of the first diffusion control section **11**, the second diffusion control section **13**, and the third diffusion control section **30** is provided as two laterally long slits (openings of which the longitudinal direction is a direction perpendicular to the drawing). The fourth diffusion control section **60** is provided as a single laterally long slit (an opening of which the longitudinal direction is a direction perpendicular to the drawing) formed as a clearance from the under surface of the

second solid electrolyte layer **6**. A part from the gas inlet port **10** to the third internal cavity **61** is also referred to as measurement-object gas flow portion.

[0037] At a location farther from the tip end side than the measurement-object gas flow portion, a reference gas inlet space **43** is provided between the top surface of the third substrate layer **3** and the under surface of the spacer layer **5** at a location at which the side part is defined by the side surface of the first solid electrolyte layer **4**. For example, the atmosphere is introduced into the reference gas inlet space **43** as a reference gas at the time of measuring a NO_x concentration.

[0038] A reference gas inlet layer **48** is a layer made of porous ceramics. The reference gas is introduced into the reference gas inlet layer **48** through the reference gas inlet space **43**. The reference gas inlet layer **48** is formed so as to coat the reference electrode **42**.

[0039] The reference electrode **42** is an electrode formed in such a manner in which the reference electrode **42** is sandwiched by the top surface of the third substrate layer **3** and the first solid electrolyte layer **4**. As described above, the reference gas inlet layer **48** that communicates with the reference gas inlet space **43** is provided around the reference electrode **42**. As will be described later, it is possible to measure an oxygen concentration (oxygen partial pressure) in the first internal cavity **20**, an oxygen concentration (oxygen partial pressure) in the second internal cavity **40**, and an oxygen concentration (oxygen partial pressure) in the third internal cavity **61** by using the reference electrode **42**. The reference electrode **42** is formed as a porous cermet electrode (for example, a cermet electrode of Pt and ZrO₂).

[0040] In the measurement-object gas flow portion, the gas inlet port **10** is a portion that is open to an external space, and a measurement-object gas is taken into the sensor element **101** from the external space through the gas inlet port **10**. The first diffusion control section **11** is a portion that applies predetermined diffusion resistance to a measurement-object gas taken in through the gas inlet port **10**. The buffer space **12** is a space provided to guide the measurement-object gas introduced from the first diffusion control section **11** to the second diffusion control section **13**. The second diffusion control section **13** is a portion that applies predetermined diffusion resistance to the measurement-object gas introduced from the buffer space **12** into the first internal cavity **20**. When the measurement-object gas is introduced from the outside of the sensor element **101** into the first internal cavity **20**, the measurement-object gas rapidly taken into the sensor element **101** through the gas inlet port **10** due to pressure fluctuations of the measurement-object gas in the external space (due to pulsation of exhaust pressure when the measurement-object gas is the exhaust gas of an automobile) is not directly introduced into the first internal cavity **20** but, after pressure fluctuations of the measurement-object gas are cancelled out through the first diffusion control section **11**, the buffer space **12**, and the second diffusion control section **13**, the measurement-object gas is introduced into the first internal cavity **20**. With this configuration, pressure fluctuations of the measurement-object gas introduced into the first internal cavity **20** are almost ignorable. The first internal cavity **20** is provided as a space used to adjust an oxygen partial pressure in the measurement-object gas introduced through the second diffusion control section **13**. The oxygen partial pressure is adjusted by the operation of a main pump cell **21**.

[0041] The main pump cell **21** is an electrochemical pump cell made up of an inner pump electrode **22** having a ceiling electrode portion **22a** provided almost all over the under surface of the second solid electrolyte layer **6**, facing the first internal cavity **20**, the outer pump electrode **23** provided so as to be exposed to the external space in a region of the top surface of the second solid electrolyte layer **6**, corresponding to the ceiling electrode portion **22a**, and the second solid electrolyte layer **6** sandwiched by these electrodes.

[0042] The inner pump electrode **22** is formed over the upper and lower solid electrolyte layers (the second solid electrolyte layer **6** and the first solid electrolyte layer **4**) defining the first internal cavity **20**, and the spacer layer **5** providing a side wall. Specifically, the ceiling electrode portion **22a** is formed on the under surface of the second solid electrolyte layer **6**, providing a ceiling surface of the first internal cavity **20**, a bottom electrode portion **22b** is formed on the top surface of

the first solid electrolyte layer **4**, providing a bottom surface, a side electrode portion (not shown) is formed on the side wall surface (inner surface) of the spacer layer **5**, making both side wall portions of the first internal cavity **20**, so as to connect those ceiling electrode portion **22a** and the bottom electrode portion **22b**, and the inner pump electrode **22** is disposed with a structure in a tunnel form at a portion where the side electrode portion is disposed.

[0043] The inner pump electrode **22** and the outer pump electrode **23** each are formed as a porous cermet electrode (for example, a cermet electrode of Pt and ZrO.sub.2, having an Au content of 1 percent). The inner pump electrode **22** that contacts with a measurement-object gas is formed by using a material of which the reduction ability for NOx components in the measurement-object gas is lowered.

[0044] By passing a pump current I_{p0} in a positive direction or a negative direction between the inner pump electrode **22** and the outer pump electrode **23** by applying a desired pump voltage V_{p0} between the inner pump electrode **22** and the outer pump electrode **23**, the main pump cell **21** is capable of pumping out oxygen in the first internal cavity **20** to the external space or pumping oxygen in the external space into the first internal cavity **20**.

[0045] In order to detect an oxygen concentration (oxygen partial pressure) in an atmosphere in the first internal cavity **20**, an electrochemical sensor cell, that is, a main pump control oxygen partial pressure detection sensor cell **80**, is made up of the inner pump electrode **22**, the second solid electrolyte layer **6**, the spacer layer **5**, the first solid electrolyte layer **4**, the third substrate layer **3**, and the reference electrode **42**.

[0046] An oxygen concentration (oxygen partial pressure) in the first internal cavity **20** is found by measuring an electromotive force (voltage V_0) in the main pump control oxygen partial pressure detection sensor cell **80**. In addition, the pump current I_{p0} is controlled by executing feedback control over the pump voltage V_{p0} of a variable power source **24** such that the voltage V_0 becomes a target value. With this configuration, it is possible to maintain the oxygen concentration in the first internal cavity **20** at a predetermined constant value.

[0047] The third diffusion control section **30** is a portion that applies predetermined diffusion resistance to a measurement-object gas of which the oxygen concentration (oxygen partial pressure) is controlled by operation of the main pump cell **21** in the first internal cavity **20** to guide the measurement-object gas to the second internal cavity **40**.

[0048] The second internal cavity **40** is provided as a space used to further adjust the oxygen partial pressure by using an auxiliary pump cell **50** for the measurement-object gas adjusted in the oxygen concentration (oxygen partial pressure) in the first internal cavity **20** in advance and then introduced through the third diffusion control section **30**. With this configuration, it is possible to highly accurately maintain the oxygen concentration in the second internal cavity **40** at a constant value, so it is possible to measure a highly accurate NOx concentration with the gas sensor **100**.

[0049] The auxiliary pump cell **50** is an auxiliary electrochemical pump cell made up of an auxiliary pump electrode **51** having a ceiling electrode portion **51a** provided substantially all over the under surface of the second solid electrolyte layer **6**, facing the second internal cavity **40**, the outer pump electrode **23** (not limited to the outer pump electrode **23**, and an adequate electrode outside the sensor element **101** may be used), and the second solid electrolyte layer **6**.

[0050] The auxiliary pump electrode **51** is disposed in the second internal cavity **40** with a structure in a similar tunnel form to that of the inner pump electrode **22** provided in the above-described first internal cavity **20**. In other words, the auxiliary pump electrode **51** has such a structure in a tunnel form that a ceiling electrode portion **51a** is formed on the second solid electrolyte layer **6** providing the ceiling surface of the second internal cavity **40**, a bottom electrode portion **51b** is formed on the first solid electrolyte layer **4** providing the bottom surface of the second internal cavity **40**, a side electrode portion (not shown) that couples those ceiling electrode portion **51a** and bottom electrode portion **51b** is formed on each of both wall surfaces of the spacer layer **5**, providing a side wall of the second internal cavity **40**. The auxiliary pump electrode **51**, as well as the inner pump electrode

22, is formed by using a material of which the reduction ability for NO_x components in the measurement-object gas is lowered.

[0051] By applying a desired voltage V_{p1} between the auxiliary pump electrode 51 and the outer pump electrode 23, the auxiliary pump cell 50 is capable of pumping out oxygen in an atmosphere in the second internal cavity 40 to the external space or pumping oxygen from the external space into the second internal cavity 40.

[0052] In order to control an oxygen partial pressure in an atmosphere in the second internal cavity 40, an electrochemical sensor cell, that is, an auxiliary pump control oxygen partial pressure detection sensor cell 81, is made up of the auxiliary pump electrode 51, the reference electrode 42, the second solid electrolyte layer 6, the spacer layer 5, the first solid electrolyte layer 4, and the third substrate layer 3.

[0053] The auxiliary pump cell 50 performs pumping with a variable power source 52 of which the voltage is controlled in accordance with an electromotive force (voltage V_1) detected by the auxiliary pump control oxygen partial pressure detection sensor cell 81. With this configuration, the oxygen partial pressure in an atmosphere in the second internal cavity 40 is controlled to a low partial pressure that substantially does not influence measurement of NO_x.

[0054] Together with this, its pump current I_{p1} is used to control the electromotive force of the main pump control oxygen partial pressure detection sensor cell 80. Specifically, the pump current I_{p1} is input to the main pump control oxygen partial pressure detection sensor cell 80 as a control signal, and the gradient of the oxygen partial pressure in the measurement-object gas to be introduced from the third diffusion control section 30 into the second internal cavity 40 is controlled to be constantly unchanged by controlling the above-described target value of the voltage V_0 . When used as a NO_x sensor, the oxygen concentration in the second internal cavity 40 is maintained at a constant value of about 0.001 ppm by the functions of the main pump cell 21 and auxiliary pump cell 50.

[0055] The fourth diffusion control section 60 is a portion that applies predetermined diffusion resistance to measurement-object gas of which the oxygen concentration (oxygen partial pressure) is controlled by operation of the auxiliary pump cell 50 in the second internal cavity 40 to guide the measurement-object gas to the third internal cavity 61. The fourth diffusion control section 60 plays a role in limiting the amount of NO_x flowing into the third internal cavity 61.

[0056] The third internal cavity 61 is provided as a space used to perform a process related to measurement of a nitrogen oxide (NO_x) concentration in a measurement-object gas on the measurement-object gas adjusted in oxygen concentration (oxygen partial pressure) in the second internal cavity 40 in advance and then introduced through the fourth diffusion control section 60. Measurement of a NO_x concentration is mainly performed by operation of a measurement pump cell 41 in the third internal cavity 61.

[0057] The measurement pump cell 41 measures a NO_x concentration in the measurement-object gas in the third internal cavity 61. The measurement pump cell 41 is an electrochemical pump cell made up of a measurement electrode 44 provided on the top surface of the first solid electrolyte layer 4, facing the third internal cavity 61, the outer pump electrode 23, the second solid electrolyte layer 6, the spacer layer 5, and the first solid electrolyte layer 4. The measurement electrode 44 is a porous cermet electrode made of a material of which the reduction ability for NO_x components in the measurement-object gas is raised as compared to the inner pump electrode 22. The measurement electrode 44 also functions as a NO_x reduction catalyst that reduces NO_x present in an atmosphere in the third internal cavity 61.

[0058] The measurement pump cell 41 is capable of pumping out oxygen produced as a result of decomposition of nitrogen oxides in an atmosphere around the measurement electrode 44 and detecting the amount of oxygen produced as a pump current I_{p2} .

[0059] In order to detect an oxygen partial pressure around the measurement electrode 44, an electrochemical sensor cell, that is, a measurement pump control oxygen partial pressure detection

sensor cell **82**, is made up of the first solid electrolyte layer **4**, the third substrate layer **3**, the measurement electrode **44**, and the reference electrode **42**. A variable power source **46** is controlled in accordance with an electromotive force (voltage V_2) detected by the measurement pump control oxygen partial pressure detection sensor cell **82**.

[0060] A measurement-object gas guided into the second internal cavity **40** reaches the measurement electrode **44** in the third internal cavity **61** through the fourth diffusion control section **60** in a situation in which the oxygen partial pressure is controlled. Nitrogen oxides in the measurement-object gas around the measurement electrode **44** are reduced ($2\text{NO} \rightarrow \text{N}_2 + \text{O}_2$) to produce oxygen. The produced oxygen is to be pumped by the measurement pump cell **41**. At this time, the voltage V_{p2} of the variable power source **46** is controlled such that the voltage V_2 detected by the measurement pump control oxygen partial pressure detection sensor cell **82** is constant (target value). The amount of oxygen produced around the measurement electrode **44** is proportional to the concentration of nitrogen oxides in the measurement-object gas, so a nitrogen oxide concentration in the measurement-object gas is calculated by using the pump current I_{p2} in the measurement pump cell **41**.

[0061] When an oxygen partial pressure detection device is constructed as an electrochemical sensor cell by combining the measurement electrode **44**, the first solid electrolyte layer **4**, the third substrate layer **3**, and the reference electrode **42**, an electromotive force according to the difference between the amount of oxygen produced by reduction of the NO_x component in the atmosphere around the measurement electrode **44**, and the amount of oxygen contained in the reference atmosphere can be detected, and accordingly, the concentration of the NO_x component in the measurement-object gas can be determined.

[0062] An electrochemical sensor cell **83** is made up of the second solid electrolyte layer **6**, the spacer layer **5**, the first solid electrolyte layer **4**, the third substrate layer **3**, the outer pump electrode **23**, and the reference electrode **42**, and it is possible to detect an oxygen partial pressure in a measurement-object gas outside the sensor by using an electromotive force (voltage V_{ref}) obtained by the sensor cell **83**.

[0063] In the gas sensor **100** having such a configuration, a measurement-object gas of which the oxygen partial pressure is maintained at a constantly unchanged low value (a value that substantially does not influence measurement of NO_x) is supplied to the measurement pump cell **41** by operating the main pump cell **21** and the auxiliary pump cell **50**. Therefore, it is possible to find a NO_x concentration in the measurement-object gas in accordance with a pump current I_{p2} that flows as a result of pumping out oxygen, produced by reduction of NO_x , by the measurement pump cell **41** substantially in proportion to a NO_x concentration in the measurement-object gas.

[0064] In addition, the sensor element **101** includes the heater section **70** that plays a role in temperature adjustment for maintaining the temperature of the sensor element **101** by heating in order to increase the oxygen ion conductivity of the solid electrolyte. The heater section **70** includes a heater connector electrode **71**, a heater **72**, a through-hole **73**, a heater insulating layer **74**, and a pressure release hole **75**.

[0065] The heater connector electrode **71** is an electrode formed in such a manner as to be in contact with the under surface of the first substrate layer **1**. Connection of the heater connector electrode **71** to an external power source allows electric power to be supplied from the outside to the heater section **70**.

[0066] The heater **72** is an electrical resistor which is formed to be vertically sandwiched between the second substrate layer **2** and the third substrate layer **3**. The heater **72** is coupled to the heater connector electrode **71** via the through-hole **73**, receives electric power supplied from the outside through the heater connector electrode **71** to generate heat, and performs heating and heat retention on the solid electrolyte constituting the sensor element **101**.

[0067] The heater **72** is embedded all over the region from the first internal cavity **20** to the third internal cavity **61**, and is capable of adjusting the overall sensor element **101** to a temperature at

which the solid electrolyte is activated.

[0068] The heater insulating layer **74** is an electrically insulating layer formed of an insulating material, such as alumina, on the top and under surfaces of the heater **72**. The heater insulating layer **74** is formed for the purpose of obtaining an electrical insulation property between the second substrate layer **2** and the heater **72** and an electrical insulation property between the third substrate layer **3** and the heater **72**.

[0069] The pressure release hole **75** is a portion provided so as to extend through the third substrate layer **3** and the reference gas inlet layer **48** and communicate with the reference gas inlet space **43**. The pressure release hole **75** is formed for the purpose of easing an increase in internal pressure resulting from an increase in temperature in the heater insulating layer **74**.

[0070] As shown in FIG. **2**, the control device **95** includes the above-mentioned variable power sources **24**, **46**, **52**, and a controller **96**. The controller **96** is a microprocessor including a CPU **97**, and a storage unit **98**. The storage unit **98** is a non-volatile memory capable of rewriting information, and can store various programs and various data, for example. The controller **96** receives input of voltage V_0 detected by the main pump control oxygen partial pressure detection sensor cell **80**, voltage V_1 detected by the auxiliary pump control oxygen partial pressure detection sensor cell **81**, voltage V_2 detected by the measurement pump control oxygen partial pressure detection sensor cell **82**, voltage V_{ref} detected by the sensor cell **83**, pump current I_{p0} detected by the main pump cell **21**, pump current I_{p1} detected by the auxiliary pump cell **50** and pump current I_{p2} detected by the measurement pump cell **41**. The controller **96** controls the voltages V_{p0} , V_{p1} , V_{p2} output by the variable power sources **24**, **46**, **52** by outputting a control signal to the variable power sources **24**, **46**, **52**, thereby controlling the main pump cell **21**, the measurement pump cell **41** and the auxiliary pump cell **50**. The storage unit **98** also stores the later-described target values V_0^* , V_1^* , V_2^* , etc. The CPU **97** of the controller **96** controls the cells **21**, **41**, **50** by referring to these target values V_0^* , V_1^* , V_2^* .

[0071] The controller **96** executes an auxiliary pump control process of controlling the auxiliary pump cell **50** so that the oxygen concentration in the second internal cavity **40** reaches a target concentration. Specifically, the controller **96** controls the auxiliary pump cell **50** by executing feedback control on the voltage V_{p1} of the variable power source **52** so that the voltage V_1 reaches a constant value (referred to as target value V_1^*). The target value V_1^* is defined as the value that causes the oxygen concentration in the second internal cavity **40** to reach a predetermined low oxygen concentration that does not substantially affect measurement of NO_x .

[0072] The controller **96** executes a main pump control process of controlling the main pump cell **21** so that the pump current I_{p1} flowing when the oxygen concentration in the second internal cavity **40** is adjusted by the auxiliary pump cell **50** in the auxiliary pump control process reaches a target current (referred to as target value I_{p1}^*). Specifically, the controller **96** sets (feedback-controls) a target value (referred to as a target value V_0^*) of the voltage V_0 based on the pump current I_{p1} so that the pump current I_{p1} caused to flow by the voltage V_{p1} reaches the constant target current I_{p1}^* . The controller **96** then performs feedback control on the pump voltage V_{p0} of the variable power source **24** so that the voltage V_0 reaches the target value V_0^* (in other words, the oxygen concentration in the first internal cavity **20** reaches the target concentration). The gradient of oxygen partial pressure in the measurement-object gas to be introduced from the third diffusion control section **30** into the second internal cavity **40** is made unchanged constantly by the main pump control process. The target value V_0^* is set to a value which causes the oxygen concentration in the first internal cavity **20** to be higher than 0% and reach a low oxygen concentration. The pump current I_{p0} which flows during the main pump control process varies according to the oxygen concentration in the measurement-object gas (that is, the measurement-object gas in the vicinity of the sensor element **101**) which flows into the measurement-object gas flow portion through the gas inlet port **10**. Thus, the controller **96** can also detect the oxygen concentration in the measurement-object gas based on the pump current I_{p0} .

[0073] The main pump control process and the auxiliary pump control process described above are also collectively referred as an adjustment pump control process. The first internal cavity **20** and the second internal cavity **40** are also collectively referred as an oxygen concentration adjustment chamber. The main pump cell **21** and the auxiliary pump cell **50** are also collectively referred as an adjustment pump cell. The controller **96** executes the adjustment pump control process, thus the adjustment pump cell adjusts the oxygen concentration in the oxygen concentration adjustment chamber.

[0074] In addition, the controller **96** performs a measurement-pump control process for controlling the measurement pump cell **41** so that the voltage V_2 reaches a constant value (referred to as a target value V_2^*) (in other words, so that the oxygen concentration in the third internal cavity **61** reaches a predetermined low concentration). Specifically, the controller **96** controls the measurement pump cell **41** by feedback-controlling the voltage V_{p2} of the variable power source **46** so that the voltage V_2 reaches the target value V_2^* . Oxygen is pumped out from the third internal cavity **61** by the measurement-pump control process. The target value V_2^* may be e.g., 380 mV or higher and 420 mV or lower.

[0075] Execution of the measurement pump control process causes oxygen to be pumped out from the third internal cavity **61** so that the oxygen produced due to reduction of NO_x in the measurement-object gas in the third internal cavity **61** become substantially zero. The controller **96** obtains a pump current I_{p2} as a detected value corresponding to the oxygen produced in the third internal cavity **61** from a specific gas (herein, NO_x), and calculates the NO_x concentration in the measurement-object gas based on the pump current I_{p2} .

[0076] The storage unit **98** stores a relational expression (for example, an expression of a linear function or a quadratic function) or a map as a correspondence relationship between the pump current I_{p2} and the NO_x concentration. Such a relational expression or map can be determined in advance by an experiment.

[0077] An example of an NO_x concentration detection process of detecting an NO_x concentration in the measurement-object gas performed by the controller **96** of thus configured the gas sensor **100** will be described below. Before starting the NO_x concentration detection process, the CPU **97** of the controller **96** first controls the electric power supplied to the heater section **70** by a heater power source which is not illustrated so that the temperature of the heater **72** reaches a target temperature (e.g., 800° C.). When the temperature of the heater **72** reaches close to the target temperature, the CPU **97** starts the NO_x concentration detection process. In the NO_x concentration detection process, the CPU **97** first starts acquisition of the respective voltages V_0 , V_1 , V_2 , V_{ref} from the above-described sensor cells **80** to **83**, and control of the above-described pump cells **21**, **41**, **50**, in other words, the adjustment-pump control process and the measurement-pump control process. When the measurement-object gas is introduced through the gas inlet **10** in this state, the measurement-object gas passes through the first diffusion control section **11**, the buffer space **12** and the second diffusion control section **13** in that order, and reaches the first internal cavity **20**. Next, in the first internal cavity **20** and the second internal cavity **40**, the oxygen concentration in the measurement-object gas is adjusted by the main pump cell **21** and the auxiliary pump cell **50**, and the measurement-object gas after the adjustment reaches the third internal cavity **61**. The CPU **97** then detects the NO_x concentration in the measurement-object gas based on the acquired pump current I_{p2} and the corresponding relationship stored in the storage unit **98**. The CPU **97** transmits the value of the detected NO_x concentration to an engine ECU which is not illustrated, and ends the NO_x concentration detection process. The CPU **97** may perform the NO_x concentration detection process e.g., at a timing for every predetermined time interval, or a timing of instructions from the engine ECU for detection of NO_x concentration.

[0078] The inventors have studied the relationship between the target value V_1^* of the gas sensor **100**, and the variation of the pump current I_{p2} due to a variation of the water concentration in the measurement-object gas. As the measurement-object gas, two types of gas, that is, a first gas and a

second gas are prepared, the first gas being a nitrogen-based gas with an oxygen concentration of 20.5%, a water concentration of 12%, and an NO concentration of 0 ppm, the second gas having the same composition of the first gas except for a water concentration of 3%. The target value $V2^*$ is set to 400 mV, the target value $V1^*$ is set to 350 mV, and the pump current I_{p2} is measured when the controller **96** performs the above-described NO_x concentration detection process on each of the first gas and the second gas. Because the first gas and the second gas each have an NO concentration of 0 ppm, the pump current I_{p2} is theoretically 0 μ A in either case; however, in actuality, a slight pump current I_{p2} flows. Such a pump current I_{p2} which flows due to a factor other than a specific gas (NO_x in this case) is referred to as an offset current $I_{p2offset}$. The value obtained by subtracting the pump current I_{p2} when the second gas is used as the measurement-object gas from the pump current I_{p2} when the first gas is used as the measurement-object gas is calculated as the $\Delta I_{p2offset}$. The $\Delta I_{p2offset}$ corresponds to the amount of change (difference) in the offset current $I_{p2offset}$ between 12% water concentration and 3% water concentration in the measurement-object gas. Under the same conditions as described above other than a changed target value $V1^*$, measurement of the pump current I_{p2} and calculation of the $\Delta I_{p2offset}$ were performed. The result is shown in FIGS. 3 and 4. FIG. 3 is a graph illustrating a relationship between the target value $V1^*$, the water concentration and the pump current I_{p2} . FIG. 4 is a graph illustrating a relationship between the target value $V1^*$ and the $\Delta I_{p2offset}$.

[0079] In FIG. 3, the relationship between the target value $V1^*$ and the pump current I_{p2} when the measurement-object gas (the first gas with a water concentration of 12% in this case) with a high water concentration is used is shown by a solid line graph, and the relationship between the target value $V1^*$ and the pump current I_{p2} when the measurement-object gas (the second gas with a water concentration of 3% in this case) with a low water concentration is used is shown by a dashed line graph. Since the NO concentration of the first gas and the second gas is 0 ppm as described above, the pump current I_{p2} in FIG. 3 is the offset current $I_{p2offset}$. In FIG. 3, the difference between the pump current I_{p2} in the solid line graph and the pump current I_{p2} in the dashed line graph both corresponding to the same target value $V1^*$ is the $\Delta I_{p2offset}$.

[0080] FIG. 4 is a graph of the relationship between the target value $V1^*$ and the $\Delta I_{p2offset}$ illustrated in FIG. 3. In FIG. 4, the correspondence between the target value $V1^*$ and the $\Delta I_{p2offset}$ for each of the target values $V1^*$ at measured eight points is shown by a black dot. FIG. 4 also shows an approximate straight line which is calculated by linear approximation based on the data of the measured eight points. The $\Delta I_{p2offset}$ along the vertical axis of FIG. 4 is shown as the value (unit is ppm) obtained by converting the above-mentioned $\Delta I_{p2offset}$ (unit is μ A) to its equivalent in terms of NO concentration using the above-mentioned corresponding relationship between the pump current I_{p2} and the NO_x concentration. In FIG. 4, for example when the target value $V1^*$ is set to 350 mV, the $\Delta I_{p2offset}$ is 5.7 ppm. This result indicates that even with the same NO_x concentration in the measurement-object gas, the values of NO_x concentration calculated based on the pump current I_{p2} for the water concentrations of 12% and 3% differ only by 5.7 ppm.

[0081] As illustrated in FIG. 3, regardless of whether the water concentration is high or low, the offset current $I_{p2offset}$ tends to be smaller (negative correlation) for a higher target value $V1^*$, and a linear relationship is observed between the target value $V1^*$ and the offset current $I_{p2offset}$. However, when the water concentration varies, the slope of the straight line showing the relationship between the target value $V1^*$ and the offset current $I_{p2offset}$ varies, and the absolute value of the negative slope of the straight line is greater for a higher water concentration. Thus, as illustrated in FIG. 3, it has been found that an intersection point is present between the solid line graph and the dashed line graph, and even if the water concentration varies, the offset current $I_{p2offset}$ does not vary at the intersection point. In other words, as illustrated in FIGS. 3 and 4, it has been found that value P1 of the target value $V1^*$ is present, at which the $\Delta I_{p2offset}$ has value 0. The value P1 calculated based on the approximate straight line of FIG. 4 was approximately 392 mV. Like the value P1, the target value $V1^*$ with the $\Delta I_{p2offset}$ of 0 ppm is referred to as an

optimal point. When the target value $V1^*$ is set to an optimal point or a value close to an optimal point, even if the water concentration in the measurement-object gas varies, the offset current $Ip2_{offset}$ is unlikely to vary, thus it is possible to prevent reduction in the accuracy of measurement due to a variation of the water concentration. This is the knowledge newly found by the inventors. Even if the target value $V1^*$ is at an optimal point (value P1) as illustrated in FIG. 3, the value of the offset current $Ip2_{offset}$ may not be 0. This is because some offset currents are also generated due to a reason other than the water concentration.

[0082] The reason why there is a relationship between the target value $V1^*$ and the $\Delta Ip2_{offset}$ as illustrated in FIGS. 3 and 4 is probably as follows.

[0083] First, the relationship between the water in the measurement-object gas and the offset current $Ip2_{offset}$ will be described. When the adjustment-pump control process (the main-pump control process and the auxiliary pump control process in this case) is performed in the presence of water in the measurement-object gas, at least part of the water is decomposed in the vicinity of the auxiliary pump electrode 51 due to the voltage $Vp1$ of the variable power source 52, and hydrogen (H.sub.2) and oxygen (O.sub.2) are produced. The produced oxygen is pumped out from the vicinity of the auxiliary pump electrode 51, in other words, from the second internal cavity 40 by the auxiliary pump cell 50; however, at least part of the produced hydrogen reaches the third internal cavity 61. The hydrogen which has reached the third internal cavity 61 reacts with the oxygen in the third internal cavity 61 to produce water, thus the amount of oxygen pumped out from the third internal cavity 61 by the measurement-pump control process decreases, in other words, the pump current $Ip2$ decreases. In contrast, when the measurement-pump control process is performed in the presence of water in the measurement-object gas, at least part of the water is decomposed in the vicinity of the measurement electrode 44 in the third internal cavity 61 due to the voltage $Vp2$ of the variable power source 46, and hydrogen (H.sub.2) and oxygen (O.sub.2) are produced. Thus, the amount of oxygen pumped out from the third internal cavity 61 in the measurement-pump control process increases, in other words, the pump current $Ip2$ increases. In this manner, the pump current $Ip2$ decreases due to the hydrogen produced from water in the vicinity of the auxiliary pump electrode 51, and the pump current $Ip2$ increases due to the oxygen produced from water in the vicinity of the measurement electrode 44, thus the sum of the amounts of decrease and increase in the pump current $Ip2$ not due to NOx in the measurement-object gas appears as the offset current $Ip2_{offset}$.

[0084] For a higher target value $V1^*$, in other words, for a lower target value of the oxygen concentration in the second internal cavity 40, the voltage $Vp1$ is likely to be a high value by the auxiliary-pump control process, thus in the vicinity of the auxiliary pump electrode 51, the amount of hydrogen produced by decomposition of water by the voltage $Vp1$ increases, thus the amount of decrease in the pump current $Ip2$ also increases. Thus, with a target value $V1^*$ cancelling out the amounts of decrease and increase in the pump current $Ip2$ due to the above-mentioned decomposition of water in the measurement-object gas, the offset current $Ip2_{offset}$ due to the decomposition of water is probably zero. When the offset current $Ip2_{offset}$ due to the decomposition of water is zero, even if the water concentration varies, the offset current $Ip2_{offset}$ does not vary, thus the $\Delta Ip2_{offset}$ is 0 μA (=0 ppm). Because of this reason, when the target value $V1^*$ is the value P1, the $\Delta Ip2_{offset}$ is 0 ppm in FIG. 4. When the target value $V1^*$ is lower than the value P1, the amount of increase in the pump current $Ip2$ due to the decomposition of water is greater than the amount of decrease in the pump current $Ip2$, thus the offset current $Ip2_{offset}$ due to decomposition of water probably has a positive value. Thus, when the target value $V1^*$ is lower than the value P1, as illustrated in FIG. 3, the pump current $Ip2$ is greater for a higher water concentration, and the $\Delta Ip2_{offset}$ probably has a positive value as illustrated in FIG. 4. In contrast, when the target value $V1^*$ is higher than the value P1, the amount of decrease in the pump current $Ip2$ due to the decomposition of water is greater than the amount of increase in the pump current $Ip2$, thus the offset current $Ip2_{offset}$ due to the decomposition of water probably has a negative

value. Thus, when the target value $V1^*$ is higher than the value $P1$, as illustrated in FIG. 3, the pump current $Ip2$ is lower for a higher water concentration, and the $\Delta Ip2offset$ probably has a negative value as illustrated in FIG. 4.

[0085] In addition, the inventors have found that an adjustment voltage target value (an optimal point or a value close to an optimal point of the target value $V1^*$ in this case) capable of preventing reduction of the accuracy of measurement due to a variation of the water concentration changes by the balance between a diffusion resistance A and a diffusion resistance B, the diffusion resistance A being the diffusion resistance from the outside of the sensor element **101** to the oxygen concentration adjustment chamber, the diffusion resistance B being the diffusion resistance from the outside of the sensor element **101** to the measurement chamber (the third internal cavity **61** in this case). More specifically, the inventors have found that the smaller the diffusion resistance B relative to the diffusion resistance A in the sensor element **101**, the optimal point for the target value $V1^*$ tends to be lower. The reason is probably as follows. Not all the hydrogen produced by the decomposition of water in the vicinity of the auxiliary pump electrode **51** reaches the third internal cavity **61** downstream of the auxiliary pump electrode **51**, and part of the hydrogen moves upstream of the auxiliary pump electrode **51**, and diffuses from the gas inlet **10** to the outside of the sensor element **101**. Thus, the proportion of degree of downstream and upstream movement (diffusion) of the hydrogen produced in the vicinity of the auxiliary pump electrode **51** relative to the measurement-object gas flow portion is changed by the diffusion resistance upstream and the diffusion resistance downstream of the auxiliary pump electrode **51**. The diffusion resistance upstream of the auxiliary pump electrode **51** is correlated with the diffusion resistance A, and the composite resistance of the diffusion resistance upstream and the diffusion resistance downstream of the auxiliary pump electrode **51** is correlated with the diffusion resistance B. In the sensor element **101** with a lower diffusion resistance B relative to the diffusion resistance A, the hydrogen produced in the vicinity of the auxiliary pump electrode **51** is more likely to move downstream, thus the amount of decrease in the pump current $Ip2$ due to the hydrogen also increases. For example, when two types of sensor element **101** having the same diffusion resistance A and different diffusion resistances B are compared, in the sensor element **101** with a lower diffusion resistance B, the hydrogen produced in the vicinity of the auxiliary pump electrode **51** is more likely to move downstream, thus the amount of decrease in the pump current $Ip2$ due to the hydrogen also increases. Thus, in the sensor element **101**, even with the same value of the target value $V1^*$, the smaller the diffusion resistance B relative to the diffusion resistance A, the amount of decrease in the pump current $Ip2$ increases. As a result, it is probable that the smaller the diffusion resistance B relative to the diffusion resistance A in the sensor element **101**, the optimal point (the target value $V1^*$ that can cancel out the amount of increase in the pump current $Ip2$ due to the decomposition of water in the vicinity of the measurement electrode **44**) of the target value $V1^*$ tends to be lower. The inventors have verified this fact in the following manner.

[0086] For the sensor element **101** used for measurement in FIGS. 3 and 4, the inventors measured, as a value correlated with the diffusion resistance A, a limiting current value La of the adjustment pump current which flows through the adjustment pump cell when the adjustment pump cell pumps out oxygen from the oxygen concentration adjustment chamber. More specifically, as the limiting current value La , the limiting current value of the pump current $Ip0$ was measured, which flows through the main pump cell **21** when the main pump cell **21** pumps out oxygen from the first internal cavity **20**. Specifically, let the diffusion resistance A be the diffusion resistance (more specifically, the composite resistance of the first diffusion control section **11** and the second diffusion control section **13**) from the outside of the sensor element **101** to the first internal cavity **20**, and the limiting current value of the pump current $Ip0$ was measured as the limiting current value La correlated with the diffusion resistance A. The flow rate per unit time of the measurement-object gas which is introduced from the outside of the sensor element **101** to the measurement-object gas flow portion through the gas inlet **10**, and reaches the first internal cavity **20**, is

negatively correlated with the diffusion resistance A upstream of the first internal cavity **20**, and the limiting current value I_a is positively correlated with the flow rate. Therefore, a negative correlation is observed between the diffusion resistance A and the limiting current value I_a , and for example, the limiting current value I_a tends to be a greater value for a lower diffusion resistance A. [0087] For the sensor element **101** used for measurement in FIG. 3, the inventors measured, as a value correlated with the diffusion resistance B, the limiting current value I_b of the measurement pump current which flows through the measurement pump cell when the measurement pump cell pumps out oxygen from the measurement chamber. More specifically, as the limiting current value I_b , the limiting current value of the pump current I_{p2} was measured, which flows through the measurement pump cell **41** when the measurement pump cell **41** pumps out oxygen from the third internal cavity **61**. In other words, let the diffusion resistance B be the diffusion resistance (more specifically, the composite resistance of the first diffusion control section **11**, the second diffusion control section **13**, the third diffusion control section **30**, and fourth diffusion control section **60**) from the outside of the sensor element **101** to the third internal cavity **61**, and the limiting current value of the pump current I_{p2} was measured as the limiting current value I_b correlated with the diffusion resistance B. The flow rate per unit time of the measurement-object gas which is introduced from the outside of the sensor element **101** to the measurement-object gas flow portion through the gas inlet **10**, and reaches the third internal cavity **61**, is negatively correlated with the diffusion resistance B upstream of the third internal cavity **61**, and the limiting current value I_b is positively correlated with the flow rate. Therefore, a negative correlation is observed between the diffusion resistance B and the limiting current value I_b , and for example, the limiting current value I_b tends to be a greater value for a lower diffusion resistance B.

[0088] The limiting current value I_a was measured in the following manner. First, as the measurement-object gas, a nitrogen-based gas (a gas having the same composition as that of the above-mentioned second gas at the time of measurement in FIGS. 3 and 4) with an oxygen concentration of 20.5%, a water concentration of 3%, and an NO concentration of 0 ppm was prepared. With the sensor element **101** exposed to the measurement-object gas, the heater **72** was energized to heat the sensor element **101** up to a predetermined target temperature (e.g., 800° C.). With no voltage applied to each of the variable power sources **46**, **52**, the measurement pump cell **41** and the auxiliary pump cell **50** were prohibited to be operated. In this state, the voltage V_{p0} is applied across the inner pump electrode **22** and the outer pump electrode **23** by the variable power source **24** so that oxygen is pumped out from the vicinity of the inner pump electrode **22** to the vicinity of the outer pump electrode **23**, and the pump current I_{p0} which flows between both electrodes **22**, **23** was measured. Subsequently, when the voltage V_{p0} is gradually increased, the pump current I_{p0} is also gradually increased; however, with the voltage V_{p0} higher than a certain level applied, even if the voltage V_{p0} is increased, the pump current I_{p0} is not increased, and reaches an upper limit. The upper limit of the pump current I_{p0} was measured as the limiting current value I_a .

[0089] The limiting current value I_b was measured in the following manner. First, as the measurement-object gas, a nitrogen-based gas with an oxygen concentration of 0%, a water concentration of 3%, and an NO concentration of 500 ppm was prepared. With the sensor element **101** exposed to the measurement-object gas, the heater **72** was energized to heat the sensor element **101** up to a predetermined target temperature (e.g., 800° C.). With no voltage applied to each of the variable power sources **24**, **52**, the main pump cell **21** and the auxiliary pump cell **50** were prohibited to be operated. In this state, the voltage V_{p2} is applied across the measurement electrode **44** and the outer pump electrode **23** by the variable power source **46** so that oxygen is pumped out from the vicinity of the measurement electrode **44** to the vicinity of the outer pump electrode **23**, and the pump current I_{p2} which flows between both electrodes **44**, **23** was measured. Subsequently, when the voltage V_{p2} is gradually increased, the pump current I_{p2} is also gradually increased; however, with the voltage V_{p2} higher than a certain level applied, even if the voltage V_{p2} is

increased, the pump current I_{p2} is not increased, and reaches an upper limit. The upper limit of the pump current I_{p2} was measured as the limiting current value L_b .

[0090] As a value representing the balance between the diffusion resistances A, B, the inventors devised the use of the value R calculated by the expression, $R=L_b^2/L_a$ in terms of the limiting current values L_a , L_b . As seen from the above expression, the value R is obtained by dividing the square of the limiting current value L_b by the limiting current value L_a . The larger the value R , the limiting current value L_b is greater relative to the limiting current value L_a , which indicates that the diffusion resistance B is lower relative to the diffusion resistance A. When the value R is calculated, the limiting current value L_a is expressed in the unit of [mA], and the limiting current value L_b is expressed in the unit of [μ A]. For the sensor element **101** used for the above-mentioned measurement in FIGS. 3 and 4, the value R was 0.998 (unit is [μ A.sup.2/mA]).

[0091] In addition to the sensor element **101** used for the measurement in FIGS. 3 and 4, four sensor elements **101** having the same value R of 0.998 as in the above sensor element **101** were prepared. For the four sensor elements **101**, the relationship between the target value $V1^*$ and the $\Delta I_{p2offset}$ was studied by the same method used for acquisition of data in FIGS. 3 and 4. In the same manner as the calculation of an approximate straight line in FIG. 4, an approximate straight line was calculated by linear approximation based on the relationship between the target value $V1^*$ and the $\Delta I_{p2offset}$ for these five sensor elements **101** in total. The target value $V1^*$ was calculated for each case where the $\Delta I_{p2offset}$ is 0 ppm, -1 ppm, +1 ppm, -3 ppm, or +3 ppm based on the approximate straight line. Similarly, three types of sensor element **101** having the value R of 0.818, 0.908, 1.137 were prepared with each type consisting of five sensor elements **101**. For these sensor elements **101**, the value R was adjusted to the above-mentioned values by changing the shape of the gap (slit) between the second diffusion control section **13** and the fourth diffusion control section **60**. For the five sensor elements **101** having the same value R , the target value $V1^*$ was calculated, which causes the $\Delta I_{p2offset}$ to be 0 ppm, -1 ppm, +1 ppm, -3 ppm, or +3 ppm by the same method described above. The result is shown in FIG. 5 and Table 1.

TABLE-US-00001 TABLE 1 $V1^*[mV]$ ΔI_{p2off} - ΔI_{p2off} - ΔI_{p2off} - ΔI_{p2off} - ΔI_{p2off} - R set = 0 ppm set = -1 ppm set = +1 ppm set = -3 ppm set = +3 ppm 0.818 397.1 404 390 415 378 0.908 396.1 402 389 413 375 0.998 394.4 400 387 410 371 1.137 392.4 398 385 407 367

[0092] As seen from the data of four points plotted as black dots in FIG. 5, it has been verified that a linear relationship is observed between the value R and the target value $V1^*$ (in other words, the optimal point of the target value $V1^*$) for which the $\Delta I_{p2offset}$ is 0 ppm, and the larger the value R , the optimal point of the target value $V1^*$ tends to be lower in the sensor element **101**. A large value R indicates that in the sensor element **101**, the limiting current value L_b is relatively larger with respect to the limiting current value L_a , and eventually, the diffusion resistance B is relatively lower with respect to the diffusion resistance A. Thus, as described above, it has been verified from FIG. 5 that in the sensor element **101** with the diffusion resistance B lower than the diffusion resistance A, the optimal point of the target value $V1^*$ tends to be lower. It has been verified that not only the optimal point of the target value $V1^*$, but also the target value $V1^*$ with the $\Delta I_{p2offset}$ relatively close to 0 ppm such as -1 ppm, +1 ppm, -3 ppm, and +3 ppm has a linear relationship with the value R . This is the knowledge newly found by the inventors. FIG. 5 shows, as a straight line L1, an approximate straight line calculated by linear approximation based on the relationship (the data of four points indicated by black dots) between the value R and the target value $V1^*$ for which the $\Delta I_{p2offset}$ is 0 ppm. Similarly, FIG. 5 shows, as straight lines L2 to L5, approximate straight lines each indicating a relationship between the value R and the target value $V1^*$ for which the $\Delta I_{p2offset}$ is -1 ppm, +1 ppm, -3 ppm, or +3 ppm. The linear functions for the straight lines L1 to L5 are given by the following Expressions (a) to (e).

[00004] StraightlineL1: $V1^* = -15.1 * R + 409.6$ (a)

StraightlineL2: $V1^* = -18.9 * R + 419.2$ (b) StraightlineL3: $V1^* = -16.2 * R + 403.4$ (c)

StraightlineL4: $V1^* = -25.7 * R + 436$. (d) StraightlineL5: $V1^* = -35.1 * R + 406.6$ (e)

[0093] From these results, it is found that in the gas sensor **100**, when the points obtained by plotting the correspondence between the value R and the target value $V1^*$ of the sensor element **101** in FIG. 5 are located between the straight line L4 and the straight line L5, in other words, when the points satisfy Expression (1) and Expression (2) below, the $\Delta Ip2offset$ can have a value relatively close to 0 ppm, such as a value within ± 3 ppm in the gas sensor **100**. Thus, in the gas sensor **100** satisfying Expression (1) and Expression (2) below, even if the water concentration in the measurement-object gas varies, the offset current $Ip2offset$ is unlikely to vary, thus it is possible to prevent reduction in the accuracy of measurement due to a variation of the water concentration. The coefficients (slope and intercept) of Expression (1) and Expression (2) below are the same as those of Expression (d) and Expression (e) above. Also, it is found that in the gas sensor **100**, when the points obtained by plotting the correspondence between the value R and the target value $V1^*$ of the sensor element **101** in FIG. 5 are located between the straight line L2 and the straight line L3, in other words, when the points satisfy Expression (3) and Expression (4) below, the $\Delta Ip2offset$ can have a value further closer to 0 ppm, such as a value within ± 1 ppm in the gas sensor **100**. Thus, the gas sensor **100** preferably satisfies Expression (3) and Expression (4) below, thereby making it possible to further prevent reduction in the accuracy of measurement due to a variation of the water concentration. In addition, it is found that in the gas sensor **100**, when the points obtained by plotting the correspondence between the value R and the target value $V1^*$ of the sensor element **101** in FIG. 5 are located on the straight line L1, in other words, when the points satisfy Expression (a) above, the target value $V1^*$ is the optimal point, and the $\Delta Ip2offset$ is substantially 0 ppm, thus reduction of the accuracy of measurement due to a variation of the water concentration does not substantially occur. Therefore, in the gas sensor **100**, the points obtained by plotting the correspondence between the value R and the target value $V1^*$ of the sensor element **101** in FIG. 5 are preferably close to the straight line L1 as much as possible, and more preferably located on the straight line L1.

[00005] $V1^* \leq -25.7 * R + 436$. (1) $V1^* \geq -35.1 * R + 406.6$ (2)

$V1^* \leq -18.9 * R + 419.2$ (3) $V1^* \geq -16.2 * R + 403.4$ (4)

[0094] In the present embodiment, in the gas sensor **100**, the target value $V1^*$ is determined based on the measured value R for the sensor element **101** so that the corresponding points between the value R and the target value $V1^*$ of the sensor element **101** are located on the straight line L1, in other words, so that the corresponding points satisfy Expression (a) above, and the determined target value $V1^*$ is stored in the storage unit **98**.

[0095] The corresponding relationship between the components of the present embodiment and the components of the present invention will now be clarified. A layered body in which six layers, that is, the first substrate layer **1**, the second substrate layer **2**, the third substrate layer **3**, the first solid electrolyte layer **4**, the spacer layer **5**, and the second solid electrolyte layer **6** of the present embodiment are layered in this order corresponds to an element body of the present invention; the third internal cavity **61** corresponds to a measurement chamber; the measurement electrode **44** corresponds to an inner measurement electrode; the measurement pump cell **41** corresponds to a measurement pump cell; the first internal cavity **20** and the second internal cavity **40** correspond to an oxygen concentration adjustment chamber; the inner pump electrode **22** and the auxiliary pump electrode **51** correspond to an inner adjustment electrode; the main pump cell **21** and the auxiliary pump cell **50** correspond to an adjustment pump cell; the reference electrode **42** corresponds to a reference electrode; the voltage V1 corresponds to an adjustment voltage, the target value $V1^*$ corresponds to an adjustment voltage target value; the voltage V2 corresponds to a measurement voltage; the target value $V2^*$ corresponds to a measurement voltage target value; the control device **95** corresponds to a control device; the pump current $Ip0$ corresponds to an adjustment pump current and a main pump current; and the pump current $Ip2$ corresponds to a measurement pump

current. In addition, the inner pump electrode **22** corresponds to an inner main pump electrode, and the auxiliary pump electrode **51** corresponds to an inner auxiliary pump electrode.

[0096] In the gas sensor **100** of the present embodiment described in detail above, the adjustment voltage target value (the target value $V1^*$ in this case) in the adjustment-pump control process satisfies Expression (1) and Expression (2) stated above. Thus, even if the water concentration in the measurement-object gas varies the pump current I_{p2} is unlikely to vary, thus it is possible to prevent reduction in the accuracy of measurement due to a variation of the water concentration in the measurement-object gas.

[0097] Also, with the above Expression (3) and Expression (4) satisfied by the adjustment voltage target value (the target value $V1^*$ in this case), it is possible to further prevent reduction in the accuracy of measurement due to a variation of the water concentration in the measurement-object gas.

Second Embodiment

[0098] An embodiment of a method for manufacturing a gas sensor, including the method for determining an adjustment voltage target value for control of a sensor element of the present invention will be described. A method for manufacturing the gas sensor **100** illustrated in FIGS. **1** and **2** will be described below. A method for manufacturing the gas sensor **100** of the present embodiment includes: a manufacturing process for manufacturing the sensor element **101**; a target value determination process for determining a target value $V1^*$ (an example of an adjustment voltage target value) for control of the manufactured sensor element **101**; an association process for associating the manufactured sensor element **101** with the determined target value $V1^*$; and a connection process for connecting the sensor element **101** associated with the target value $V1^*$ to the control device **95** that feedback-controls the auxiliary pump cell **50** (an example of the adjustment pump cell) so that the voltage $V1$ across the reference electrode **42** and the auxiliary pump electrode **51** (an example of the inner adjustment electrode, and an example of the inner auxiliary pump electrode) reaches the associated target value $V1^*$.

[0099] In the manufacturing process, the sensor element **101** is manufactured in the following manner, for example. First, six unfired ceramic green sheets containing, as a ceramic component, an oxygen-ion-conductive solid electrolyte such as zirconia are prepared. Multiple sheet holes and necessary through-holes and the like used for positioning at the time of printing or at the time of layering are formed in the green sheets in advance. Also, a green sheet that becomes the spacer layer **5** is provided with a space by a punching process or the like in advance, the space to become the measurement-object gas flow portion. A pattern printing process and a drying process for forming various patterns on the ceramic green sheets are performed corresponding to the first substrate layer **1**, the second substrate layer **2**, the third substrate layer **3**, the first solid electrolyte layer **4**, the spacer layer **5**, and the second solid electrolyte layer **6**. Specifically, the formed patterns are, for example, the above-mentioned electrodes, lead wires connected to the electrodes, the reference gas inlet layer **48**, and the heater section **70**. The pattern printing is performed by applying paste for pattern formation to the green sheets using a publicly known screen printing technique, the paste for pattern formation being prepared according to the characteristics required for respective objects to be formed. The drying process is also performed using a publicly known drying device. When the pattern printing and drying are completed, a printing and drying process is performed on paste for bonding for layering and bonding the green sheets corresponding to the layers. The green sheets with the paste for bonding formed undergo positioning with sheet holes, and are layered in a predetermined order, then a pressure bonding process is performed by applying predetermined temperature and pressure conditions, thereby producing one layered body. Thus obtained layered body includes a plurality of sensor elements **101**. The layered body is cut into pieces each having the size of the sensor element **101**. The cut pieces of the layered body are fired at a predetermined firing temperature to obtain the sensor elements **101**.

[0100] Note that the first to fourth diffusion control sections **11**, **13**, **30**, **60** can be formed by

printing a paste composed of a vanishing material (e.g., theobromine) on the upper and lower surfaces of the green sheet that becomes the spacer layer **5** at the time of pattern printing process, and vanishing the vanishing material at the time of firing the layered body.

[0101] Next, a target value determination process for determining a target value $V1^*$ is performed on each of multiple manufactured sensor elements **101**. The target value determination process includes a step (a) of determining the target value $V1^*$ (an example of the adjustment voltage target value) so that the lower a diffusion resistance B relative to a diffusion resistance A, the target value $V1^*$ tends to be lower, where the diffusion resistance A is a diffusion resistance from the outside of the sensor element **101** to the first internal cavity **20** (an example of the oxygen concentration adjustment chamber), and the diffusion resistance B is a diffusion resistance from the outside of the sensor element **101** to the third internal cavity **61** (an example of the measurement chamber).

[0102] As described in the first embodiment, the target value $V1^*$ (an optimal point or a value close to an optimal point of the target value $V1^*$) capable of preventing reduction of the accuracy of measurement due to a variation of the water concentration changes by the balance between the diffusion resistance A and the diffusion resistance B of the sensor element **101**. Thus, in the sensor element **101** as the object for determination of the target value $V1^*$, the target value $V1^*$ is easily determined to an optimal point or a value close to an optimal point by determining the target value $V1^*$ so that the smaller the diffusion resistance B relative to the diffusion resistance A, the target value $V1^*$ tends to be lower. Controlling the sensor element based on thus determined target value $V1^*$ can prevent reduction in the accuracy of measurement of a specific gas concentration due to a variation of the water concentration in the measurement-object gas. In the step (a), the target value $V1^*$ may be determined as the value corresponding to the diffusion resistances A, B of the sensor element **101** as the object for determination of the target value $V1^*$ based on a prepared corresponding relationship in advance between the diffusion resistances A, B and the target value $V1^*$.

[0103] The diffusion resistance A of the sensor element **101** can be calculated based on e.g., the dimensions (the dimensions of the gap, that is, the slit) of the diffusion control sections present inclusively between the outside of the sensor element **101** and the oxygen concentration adjustment chamber. The diffusion resistance can be calculated, for example, by dividing the length (the length in the front-back direction in this case) of a diffusion control section in the flow direction of the measurement-object gas by the cross-sectional area perpendicular to the flow direction. Similarly, the diffusion resistance B can be calculated based on e.g., the dimensions of the diffusion control sections present inclusively between the outside of the sensor element **101** and the measurement chamber. The diffusion resistances A, B may be calculated by measuring the dimensions of the diffusion control sections of the manufactured sensor element **101**, or calculated based on the design value of the dimensions of the diffusion control sections when the sensor element **101** is manufactured. The dimensions of the diffusion control sections are adjustable by adjusting the pattern shape at the time of printing of the above-mentioned paste composed of a vanishing material, or adjusting the thickness of the paste after printing by adjusting the viscosity of the paste. Even when the values of the diffusion resistances A, B themselves cannot be calculated, the step (a) can be performed based on the balance between the diffusion resistance A and the diffusion resistance B. For example, when two types of the sensor element **101** having the same diffusion resistance A and different diffusion resistances B are provided on determining the target values $V1^*$ (an example of the adjustment voltage target value) of multiple sensor elements **101**, for the sensor element **101** having a lower diffusion resistance B, the target value $V1^*$ may be determined to be a smaller value. In this case, as compared to when target values $V1^*$ for multiple sensor elements **101** are determined to be the same value uniformly regardless of the balance between diffusion resistance A and diffusion resistance B, the target value $V1^*$ is likely to be an optimal point or a value close to an optimal point. In the present embodiment, the composite resistance of the first diffusion control section **11** and the second diffusion control section **13** which are located upstream

of the first internal cavity **20** of the measurement-object gas flow portion corresponds to the diffusion resistance A. The composite resistance of the first diffusion control section **11**, the second diffusion control section **13**, the third diffusion control section **30**, and fourth diffusion control section **60** which are located upstream of the third internal cavity **61** of the measurement-object gas flow portion corresponds to the diffusion resistance B.

[0104] The target value $V1^*$ may be determined based on a value correlated with the diffusion resistances A, B instead of based on the diffusion resistances A, B. For example, the above-described step (a) may include: a step (a1) of measuring, as a value correlated with the diffusion resistance A, the limiting current value I_a of the pump current I_{p0} (an example of the adjustment pump current) which flows through the main pump cell **21** (an example of the adjustment pump cell) when the main pump cell **21** pumps out oxygen from the first internal cavity **20** (an example of the oxygen concentration adjustment chamber), and as a value correlated with the diffusion resistance B, the limiting current value I_b of the pump current I_{p2} (the measurement pump current) which flows through the measurement pump cell **41** when the measurement pump cell **41** pumps out oxygen from the third internal cavity **61** (an example of the measurement chamber); and a step (a2) of determining the target value $V1^*$ so that for the limiting current values I_a , I_b measured in the step (a1), the greater the limiting current value I_b relative to the limiting current value I_a , the target value $V1^*$ tends to be lower.

[0105] As described in the first embodiment, a negative correlation is observed between the diffusion resistance A and the limiting current value I_a , and a negative correlation is observed between the diffusion resistance B and the limiting current value I_b . Thus, the step (a1) and the step (a2) are performed to measure the limiting current values I_a , I_b instead of the diffusion resistances A, B, and the target value $V1^*$ can be determined to an optimal point or a value close to an optimal point based on the measured values. In the manufactured sensor element **101**, the limiting current values I_a , I_b can be measured non-destructively for the sensor element **101**. Thus, for example, as compared to when the diffusion resistances A, B calculated based on the design value are used, in the step (a1), the limiting current values I_a , I_b can be obtained as values correlated with the actual diffusion resistances A, B which include an individual difference such as a manufacturing error of the sensor element **101**. Thus, the target value $V1^*$ can be determined to be a more appropriate value.

[0106] The limiting current values I_a , I_b can be measured in the same manner as in the first embodiment. However, in the first embodiment, the measurement-object gas used to measure the limiting current value I_a is a nitrogen-based gas with an oxygen concentration of 20.5%, a water concentration of 3%, and an NO concentration of 0 ppm, and the measurement-object gas used to measure the limiting current value I_b is a nitrogen-based gas with an oxygen concentration of 0%, a water concentration of 3%, and an NO concentration of 500 ppm; however, without being limited to this, the limiting current values I_a , I_b may be measured using a gas with another composition. For example, the oxygen concentration of the measurement-object gas at the time of measurement of the limiting current value I_a is not limited to 20.5%, and it is sufficient that the measurement-object gas contain oxygen with a concentration which causes the pump current I_{p0} to be a limiting current. The measurement-object gas at the time of measurement of the limiting current value I_b may contain oxygen or oxide (e.g., NO) with a concentration which causes the pump current I_{p2} to be a limiting current. Thus, the oxygen concentration of the measurement-object gas at the time of measurement of the limiting current value I_b is not limited to 0%, and the measurement-object gas may contain oxygen. The NO concentration of the measurement-object gas at the time of measurement of the limiting current value I_a is not limited to 0 ppm, and may be a value greater than 0 ppm. The NO concentration of the measurement-object gas at the time of measurement of the limiting current value I_b is not limited to 500 ppm, and may be another value, or may be 0 ppm. When the measurement-object gas contains oxygen, even if the NO concentration is 0 ppm, the limiting current value I_b can be measured. The NO concentration of the measurement-object

gas may be the same both at the time of measurement of the limiting current value L_a and at the time of measurement of the limiting current value L_b . The water concentration of the measurement-object gas at the time of measurement of the limiting current values L_a , L_b is not limited to 3%, and may be another value. However, the water concentration of the measurement-object gas is preferably the same both at the time of measurement of the limiting current value L_a and at the time of measurement of the limiting current value L_b . The measurement-object gas at the time of measurement of the limiting current value L_a , and the measurement-object gas at the time of measurement of the limiting current value L_b may be the same gas. In the first embodiment, at the time of measurement of the limiting current value L_a , the measurement pump cell **41** and the auxiliary pump cell **50** are prohibited to be operated, but may be operated. In the first embodiment, at the time of measurement of the limiting current value L_b , the main pump cell **21** and the auxiliary pump cell **50** are prohibited to be operated, but may be operated. When the measurement-object gas at the time of measurement of the limiting current value L_b contains oxygen, the main pump cell **21** and the auxiliary pump cell **50** are preferably operated.

[0107] In the above-described step (a2), the target value $V1^*$ may be determined so that the greater the above-mentioned value $R (=L_b \cdot L_b / L_a)$, the target value $V1^*$ tends to be lower. As described in the first embodiment, the value R measured using the sensor element **101** has a linear relationship with the optimal point of the target value $V1^*$. Thus, the target value $V1^*$ is easily determined to an optimal point or a value close to an optimal point by determining the target value $V1^*$ using the value R so that the target value $V1^*$ tends to be lower for a larger value R .

[0108] In the step (a2), the target value $V1^*$ may be determined as a value corresponding to the limiting current values L_a , L_b measured in the step (a1) based on a prepared corresponding relationship in advance between the limiting current values L_a , L_b and the target value $V1^*$. In this manner, an appropriate corresponding relationship is prepared in advance, for example, by conducting experiments on multiple sensor elements **101**, thus in the step (a2), an appropriate target value $V1^*$ can be easily determined using the corresponding relationship. For example, Expression (1) to Expression (4) and Expression (a) to Expression (e) described in the first embodiment are each a corresponding relationship between the value R and preferred target value $V1^*$, thus an example of prepared corresponding relationship in advance between the limiting current values L_a , L_b and the target value $V1^*$. Thus, for example, in the step (a2), the target value $V1^*$ corresponding to the limiting current values L_a , L_b measured in the step (a1) can be determined using Expression (1) and Expression (2) as the prepared corresponding relationships in advance. Specifically, the value R calculated from the limiting current values L_a , L_b is substituted into Expression (1) and Expression (2) to calculate the numerical range of the target value $V1^*$ satisfying Expression (1) and Expression (2), and the target value $V1^*$ may be determined as the value in the numerical range. In this manner, the target value $V1^*$ satisfying Expression (1) and Expression (2), in other words, the target value $V1^*$ such that the above-mentioned the $\Delta I_{p2offset}$ is within ± 3 ppm can be determined. As the prepared corresponding relationship in advance, Expression (3) and Expression (4) may be used, or Expression (a) may be used.

[0109] Note that when a different method for measuring the limiting current values L_a , L_b is adopted, the corresponding relationship between the limiting current values L_a , L_b and a preferred target value $V1^*$ may also change, thus the corresponding relationship is not limited to the above Expression (1) to Expression (4) and Expression (a) to Expression (e). The limiting current values L_a , L_b may be measured under the same condition at the time of preparation of a corresponding relationship in advance and at the time of execution of the step (a1).

[0110] When the target value $V1^*$ is determined in the target value determination process, an association process is performed to associate the determined value with the sensor element **101** which is manufactured in the manufacturing process and serves as the object for determination of the target value $V1^*$. In other words, in each of the sensor elements **101** manufactured in the manufacturing process, the determined target value $V1^*$ is made to be identified. Direct association

may be made, for example, by transporting, along with the sensor element **101**, paper or the like on which the value of the target value $V1^*$ is written. Alternatively, indirect association may be made, for example, by storing, in a storage medium, a corresponding relationship between the manufacturing number assigned to the sensor element **101** and the value of the target value $V1^*$, or managing traceability of the manufacturing process as online data by e.g., a server.

[0111] Subsequently, in the connection process, the control device **95** is connected to the sensor element **101**. The value of the target value $V1^*$ associated with the sensor element **101** in the association process is stored in the storage unit **98**. Thus, the control device **95** can feedback-control the auxiliary pump cell **50** so that the voltage $V1$ across the reference electrode **42** and the auxiliary pump electrode **51** reaches the target value $V1^*$ determined in the target value determination process. In the connection process, connecting the control device **95** to the sensor element **101** and storing the value of the target value $V1^*$ in the storage unit **98** may be performed in any order. Thereby, the gas sensor **100** is manufactured. In the gas sensor **100**, the value of the target value $V1^*$ is the value determined in the above-mentioned target value determination process, thus reduction of the accuracy of measurement of a specific gas concentration due to a variation of the water concentration in the measurement-object gas is prevented.

[0112] The method for manufacturing a gas sensor in the second embodiment is also an example of the method for manufacturing the gas sensor **100** in the first embodiment described above, for example, the gas sensor **100** satisfying the above Expression (1) and Expression (2), or the gas sensor **100** satisfying the above Expression (3) and Expression (4). In the second embodiment, an example of the method for determining an adjustment voltage target value for control of a sensor element of the present invention will also be clarified by describing the method for manufacturing the sensor element **101** and the gas sensor **100**.

[0113] In the method for determining an adjustment voltage target value of the sensor element **101** of the present embodiment described in detail above, the adjustment voltage target value is determined so that the smaller the diffusion resistance B relative to the diffusion resistance A , the adjustment voltage target value (the target value $V1^*$ in this case) tends to be lower, thus the adjustment voltage target value is easily determined to an optimal point or a value close to an optimal point. Controlling the sensor element **101** based on thus determined adjustment voltage target value facilitates prevention of reduction in the accuracy of measurement of a specific gas concentration due to a variation of the water concentration in the measurement-object gas.

[0114] Also, the limiting current values I_a , I_b as values correlated with the diffusion resistances A , B are measured, and the adjustment voltage target value is determined so that the greater the limiting current value I_b relative to the measured limiting current value I_a , the adjustment voltage target value tends to be lower. Thus, instead of measuring the diffusion resistance A and the diffusion resistance B , the limiting current value I_a and the limiting current value I_b are measured, and the adjustment voltage target value can be determined to an optimal point or a value close to an optimal point.

[0115] In addition, the adjustment voltage target value is determined so that the greater the value R ($=I_b^2/I_a$), the adjustment voltage target value tends to be lower. Since the value R has a linear relationship with the optimal point of the adjustment voltage target value, using the value R , the adjustment voltage target value is easily determined to an optimal point or a value close to an optimal point.

[0116] The adjustment voltage target value is then determined as the value corresponding to the measured limiting current values I_a , I_b based on the prepared corresponding relationship in advance between the limiting current values I_a , I_b and the adjustment voltage target value. Thus, an appropriate adjustment voltage target value can be easily determined using the prepared corresponding relationship in advance.

Third Embodiment

[0117] Another embodiment of the method for manufacturing a gas sensor, including the method

for determining an adjustment voltage target value for control of a sensor element of the present invention will be described. The method for manufacturing a gas sensor of the present embodiment is the same as the method for manufacturing a gas sensor of the second embodiment except that the target value determination process is different. Thus, a description of any process other than the target value determination process will be omitted.

[0118] The target value determination process of the present embodiment includes: a step (b1) of setting the target value $V2^*$ (an example of the measurement voltage target value) to a predetermined value, setting the target value $V1^*$ (an example of the adjustment voltage target value) to a temporary target value, and measuring a first measurement pump current and a second measurement pump current, the first measurement pump current being the pump current I_{p2} which flows through the measurement pump cell **41** when the adjustment-pump control process and the measurement-pump control process are performed on a first gas with a water concentration of a first concentration $Ch1$ as the measurement-object gas, the second measurement pump current being the pump current I_{p2} which flows through the measurement pump cell **41** when the adjustment-pump control process and the measurement-pump control process are performed on a second gas with a water concentration of a second concentration $Ch2$ lower than the first concentration $Ch1$ as the measurement-object gas; and a step (b2) of changing the temporary target value and performing the step (b1) for multiple times, and determining a temporary target value to be the target value $V1^*$ for control of the sensor element **101**, the temporary target value causing the discrepancy between the first measurement pump current and the second measurement pump current to fall within a predetermined minute range.

[0119] As described in the first embodiment, the gas sensor element **101** has an optimal point for the target value $V1^*$, the optimal point being capable of preventing reduction in the accuracy of measurement of a specific gas concentration due to a variation of the water concentration in the measurement-object gas. Thus, when a temporary target value of the target value $V1^*$ is an optimal point or a value close to an optimal point, the first measurement pump current and the second measurement pump current measured in the step (b1) have the same value or values close to each other. Thus, an appropriate target value $V1^*$ can be found by changing the temporary target value and performing the step (b1) for multiple times. In the step (b2), in this manner, the temporary target value is changed and the step (b1) is performed for multiple times, and the temporary target value causing the discrepancy between the first measurement pump current and the second measurement pump current to fall within a predetermined minute range is determined to be the target value $V1^*$ for control of the sensor element **101**, thus the target value $V1^*$ can be determined to an optimal point or a value close to an optimal point. Controlling the sensor element **101** based on thus determined target value $V1^*$ can prevent reduction in the accuracy of measurement of a specific gas concentration due to a variation of the water concentration in the measurement-object gas. When the target value $V1^*$ is determined in this manner, it is not necessary to measure the diffusion resistances A , B or the limiting current values I_a , I_b as in the second embodiment.

[0120] A predetermined minute range can be defined in advance so that the effect of the discrepancy between the first measurement pump current and the second measurement pump current on the accuracy of measurement of a specific gas concentration is within an acceptable range. As the value representing the discrepancy between the first measurement pump current and the second measurement pump current, the above-mentioned $\Delta I_{p2offset}$, that is, the difference between the first measurement pump current and the second measurement pump current may be used. For example, in the step (b2), it is determined whether the absolute value of the $\Delta I_{p2offset}$ is less than or equal to a predetermined threshold value, (for example, whether the absolute value is less than or equal to 3 ppm, or whether the absolute value is less than or equal to 1 ppm in terms of NO concentration conversion), and when an affirmative determination is made, the temporary target value may be determined to be the target value $V1^*$ for control of the sensor element **101**. As the value representing the discrepancy between the first measurement pump current and the second

measurement pump current, the ratio of the first measurement pump current to the second measurement pump current may be used. For example, in step (b2), it is determined whether the ratio is within a predetermined range which is close to value 1, and when an affirmative determination is made, the temporary target value may be determined to the target value $V1^*$ for control of the sensor element **101**.

[0121] The first gas and the second gas used in the step (b1) may have the same composition as that of the first gas and the second gas (the measurement-object gas at the time of calculation of the $\Delta I_{p2offset}$ illustrated in FIG. 4) described in the first embodiment. However, without being limited to this, it is sufficient that the water concentration (second concentration $Ch2$) of the second gas be lower than the water concentration (first concentration $Ch1$) of the first gas. For example, the first gas and the second gas may differ in at least one of oxygen concentration and NO concentration. However, the first gas and the second gas preferably have the same composition as much as possible except for the water concentration. Thus, the first gas and the second gas preferably have the same oxygen concentration and the same NO concentration. Each of the first gas and the second gas preferably has an NO concentration of 0 ppm. In the step (b1), either one of the first measurement pump current and the second measurement pump current may be measured first.

[0122] The predetermined value of the target value $V2^*$ at the time of measurement of the first measurement pump current and the second measurement pump current in the step (b1) is preferably the same as or close to the target value $V2^*$ for the measurement-pump control process when the sensor element **101** is actually used for detection of a specific gas concentration. In the present embodiment, the predetermined value of the target value $V2^*$ is 400 mV which is the same as at the time of use of the sensor element **101**. The predetermined value may be e.g., 380 mV or higher and 420 mV or lower.

[0123] As a result, the gas sensor **100** of the above-described first embodiment, for example, the gas sensor **100** satisfying the Expression (1) and Expression (2), and the gas sensor **100** satisfying the Expression (3) and Expression (4) can be manufactured by the manufacturing method including the target value determination process of the third embodiment.

[0124] In the method for determining an adjustment voltage target value of the sensor element **101** of the present embodiment described in detail above, the temporary target value of the adjustment voltage target value (the target value $V1^*$ in this case) is changed and the step (b1) is performed for multiple times, and the temporary target value causing the discrepancy between the first measurement pump current and the second measurement pump current to fall within a predetermined minute range is determined to be the adjustment voltage target value for control of the sensor element **101**, thus the adjustment voltage target value can be determined to an optimal point or a value close to an optimal point. Controlling the sensor element **101** based on thus determined adjustment voltage target value can prevent reduction in the accuracy of measurement of a specific gas concentration due to a variation of the water concentration in the measurement-object gas.

[0125] Note that the present invention is not limited to the above-described embodiment at all, and it is needless to say that the present invention can be carried out in various forms as long as the forms belong to the technical scope of the present invention.

[0126] For example, in the above-described first to third embodiments, the oxygen concentration adjustment chamber includes the first internal cavity **20** and the second internal cavity **40**; however, without being limited to this, for example, the oxygen concentration adjustment chamber may further include another internal cavity, or one of the first internal cavity **20** and the second internal cavity **40** may be omitted. Similarly, in the above-described first to third embodiments, the adjustment pump cell includes the main pump cell **21** and the auxiliary pump cell **50**; however, without being limited to this, for example, the adjustment pump cell may further include another pump cell, or one of the main pump cell **21** and the auxiliary pump cell **50** may be omitted. For example, when the oxygen concentration of the measurement-object gas can be sufficiently

reduced only by the main pump cell **21**, the auxiliary pump cell **50** may be omitted. When the auxiliary pump cell **50** is omitted, the controller **96** only need to perform the main-pump control process as the adjustment-pump control process. In the main-pump control process, the above-described setting of the target value $V0^*$ based on the pump current I_{p1} may be omitted. Specifically, a predetermined target value $V0^*$ is pre-stored in the storage unit **98**, and the controller **96** may control the main pump cell **21** by feed-back controlling the voltage V_{p0} of the variable power source **24** so that the voltage $V0$ reaches the target value $V0^*$. When the auxiliary pump cell **50** is omitted, the target value $V0^*$ corresponds to the adjustment voltage target value (the target value $V1^*$ in the first to third embodiments). Thus, for example, $V1^*$ in Expression (1) to Expression (5) indicates the target value $V0^*$. When the oxygen concentration adjustment chamber further includes another internal cavity in addition to the first internal cavity **20** and the second internal cavity **40**, and the adjustment pump cell includes three or more pump cells, the target value of the voltage in the control process of the most downstream pump cell among the three or more pump cells, in other words, the pump cell disposed closest to the measurement chamber corresponds to the adjustment voltage target value (the target value $V1^*$ in the first to third embodiments). When the adjustment pump cell includes three or more pump cells, the limiting current value I_a to be measured in the second embodiment may be the limiting current value of any pump cell among the three or more pump cells.

[0127] In the above-described first to third embodiments, the sensor element **101** of the gas sensor **100** includes the first internal cavity **20**, the second internal cavity **40**, and the third internal cavity **61**, but is not limited thereto. For example, as the sensor element **201** of FIG. 6, the third internal cavity **61** may not be included. In the sensor element **201** of the modification illustrated in FIG. 6, the gas inlet **10**, the first diffusion control section **11**, the buffer space **12**, the second diffusion control section **13**, the first internal cavity **20**, the third diffusion control section **30**, and the second internal cavity **40** are formed adjacently in that order in a communicating manner between the lower surface of the second solid electrolyte layer **6** and the upper surface of the first solid electrolyte layer **4**. The measurement electrode **44** is disposed on the upper surface of the first solid electrolyte layer **4** in the second internal cavity **40**. The measurement electrode **44** is covered by the fourth diffusion control section **45**. The fourth diffusion control section **45** is a film made of a ceramic porous body such as alumina (Al_2O_3). As with the fourth diffusion control section **60** of the above-described embodiment, the fourth diffusion control section **45** has a function of limiting the amount of NO_x which flows into the measurement electrode **44**. The fourth diffusion control section **45** also functions as a protective film of the measurement electrode **44**. The ceiling electrode portion **51a** of the auxiliary pump electrode **51** is formed to a position immediately above the measurement electrode **44**. Even with thus configured sensor element **201**, as in the above-described embodiment, NO_x concentration can be detected, for example, based on the pump current I_{p2} . In this case, the vicinity of the measurement electrode **44** functions as a measurement chamber. Also, the composite resistance of the first diffusion control section **11**, the second diffusion control section **13**, the third diffusion control section **30** and the fourth diffusion control section **45** corresponds to the diffusion resistance B .

[0128] In the above-described second embodiment, in the step (a2) of the target value determination process, as a prepared corresponding relationship in advance between the limiting current values I_a , I_b and the target value $V1^*$, a corresponding relationship between the value R and the target value $V1^*$ has been exemplified, but is not limited thereto. For example, a corresponding relationship between the ratio ($=I_b/I_a$) of the limiting current value I_b to the limiting current value I_a , and the target value $V1^*$ may be prepared in advance. Alternatively, without using a value (such as the value R or the above-mentioned ratio) calculated from the limiting current value I_a and the limiting current value I_b , a corresponding relationship between the limiting current value I_a and the limiting current value I_b , and the target value $V1^*$ may be prepared in advance.

[0129] In the above-described second embodiment, the limiting current value I_a measured in the step (a1) is the limiting current value of the pump current I_{p0} which flows through the main pump cell **21** when the main pump cell **21** pumps out oxygen from the first internal cavity **20**, but is not limited thereto. The limiting current value I_a may be the limiting current value of the adjustment pump current which flows through the adjustment pump cell when the adjustment pump cell pumps out oxygen from the oxygen concentration adjustment chamber. For example, in the step (a1), as the limiting current value I_a , the limiting current value of the pump current I_{p1} may be measured, which flows through the auxiliary pump cell **50** when the auxiliary pump cell **50** pumps out oxygen from the second internal cavity **40**. The limiting current value of the pump current I_{p1} is correlated with the diffusion resistance from the outside of the sensor element **101** to the second internal cavity **40** (an example of the oxygen concentration adjustment chamber). Thus, as with the limiting current value of the pump current I_{p0} , the limiting current value of the pump current I_{p1} can be utilized as a value related to the degree of upstream movement (diffusion) of hydrogen relative to the measurement-object gas flow portion, the hydrogen being produced in the vicinity of the auxiliary pump electrode **51**. In this case, the composite resistance of the first diffusion control section **11**, the second diffusion control section **13** and the third diffusion control section **30** which are located upstream of the second internal cavity **40** of the measurement-object gas flow portion corresponds to the diffusion resistance A .

[0130] The target value determination process (the method for determining an adjustment voltage target value) of the second, third embodiments described above may be performed at a timing other than the time of manufacturing of a sensor element and the time of manufacturing of a gas sensor. For example, for the gas sensor **100** with an adjustment voltage target value already determined, mounted on a vehicle, the target value determination process of the second embodiment may be performed to calibrate the adjustment voltage target value. When the measurement-object gas has an oxygen concentration which allows the limiting current values I_a , I_b to be measured, an adjustment voltage target value can be determined by performing the steps (a1), (a2) described above. For example, the target value determination process may be performed at a target value determination timing at which the measurement-object gas is assumed to be an atmospheric atmosphere. For example, the target value determination timing may be a timing at which the measurement-object gas is assumed to be an exhaust gas at the time of fuel cut of an internal combustion engine. As described above, the controller **96** can detect an oxygen concentration in the measurement-object gas based on the pump current I_{p0} which is caused to flow due to the main-pump control process, thus can detect a target value determination timing based on the pump current I_{p0} . For example, when the pump current I_{p0} suddenly increases or increases to a value exceeding a predetermined threshold value, the controller **96** may determine that a target value determination timing has reached. At the target value determination timing, the controller **96** may determine an adjustment voltage target value in the following manner. First, the controller **96** stops the adjustment-pump control process and the measurement-pump control process. Next, the controller **96** performs the step (a1) to measure the limiting current value I_a and the limiting current value I_b . As in the first embodiment, the controller **96** may measure the limiting current value I_a by gradually increasing the voltage V_{p0} . Alternatively, the controller **96** may set the target value $V0^*$ to a relatively high predetermined value (a value higher than the target value $V0^*$ set in the main-pump control process at the time of measurement of NO_x concentration) which causes the pump current I_{p0} to be a limiting current, and perform the same process as the main-pump control process to measure the pump current I_{p0} which flows at this point as the limiting current value I_a . Similarly, as in the first embodiment, the controller **96** may measure the limiting current value I_b by gradually increasing the voltage V_{p2} , or may set the target value $V2^*$ to a relatively high predetermined value which causes the pump current I_{p2} to be a limiting current, and perform the measurement-pump control process to measure the pump current I_{p2} which flows at this point as the limiting current value I_b . As described above, the limiting current value of the pump current

Ip1 may be measured as the limiting current value La. The controller **96** then performs the step (a2) to determine the target value V1* based on the limiting current values La, Lb measured in the step (a1). In the step (a2), a corresponding relationship between limiting current values La, Lb, and an appropriate target value V1* is prepared in advance and stored in the storage unit **98**, the limiting current values La, Lb being measured in a state where the measurement-object gas is an exhaust gas (e.g., an atmosphere) at the time of fuel cut, then the target value V1* is preferably determined based on the stored corresponding relationship and the limiting current values La, Lb measured in the step (a1). The controller **96** calibrates the target value V1* by overwriting the currently set value of the target value V1* with thus determined target value V1*. In the sensor element **101**, the diffusion resistance of the first diffusion control section **11**, the second diffusion control section **13**, the third diffusion control section **30**, and the fourth diffusion control section **60** may vary due to use, and thus the balance between the diffusion resistances A, B may be changed. Even in this situation, an appropriate target value V1* is likely to be maintained by the controller **96** calibrating the target value V1*, thus it is possible to prevent increase of the effect of reduction in the accuracy of measurement of a specific gas concentration due to a variation of the water concentration in the measurement-object gas with use of the sensor element **101**.

[0131] In the above-described embodiment, the outer pump electrode **23** plays a role as the electrode (also referred to as an outer main pump electrode) to be paired with the inner pump electrode **22** in the main pump cell **21**, plays a role as the electrode (also referred to as an outer auxiliary pump electrode) to be paired with the auxiliary pump electrode **51** in the auxiliary pump cell **50**, and plays a role as the electrode (also referred to as an outer measurement electrode) to be paired with the measurement electrode **44** in the measurement pump cell **41**; however, the configuration is not limited thereto. One or more of the outer main pump electrode, the outer auxiliary pump electrode, and the outer measurement electrode may be provided separately from the outer pump electrode **23** outside the element body so as to be in contact with the measurement-object gas.

[0132] In the above-described embodiment, the sensor element **101** detects the NOx concentration in the measurement-object gas; however, as long as the sensor element **101** detects the concentration of a specific gas in the measurement-object gas, the configuration is not limited thereto. For example, not only NOx, but also another oxide concentration may serve as the specific gas concentration. When the specific gas is an oxide, as in the above-described embodiment, oxygen is produced when the specific gas itself is reduced in the third internal cavity **61**, thus the measurement pump cell **41** can detect the specific gas concentration by obtaining a detection value (e.g., the pump current Ip2) corresponding to the oxygen. Alternatively, the specific gas may be a non-oxide such as ammonia. When the specific gas is a non-oxide, the specific gas is converted to an oxide (for example, ammonia is converted to NO), thereby producing oxygen when the gas after the conversion is reduced in the third internal cavity **61**, thus the measurement pump cell **41** can detect the specific gas concentration by obtaining a detection value (e.g., the pump current Ip2) corresponding to the oxygen. For example, the inner pump electrode **22** in the first internal cavity **20** functions as a catalyst, thus ammonia can be converted to NO in the first internal cavity **20**. The specific gas may be a predetermined gas other than oxygen. The specific gas may be a predetermined gas other than oxygen and carbon dioxide. The specific gas may be one of NOx and ammonia.

[0133] In the above-described embodiment, the element body of the sensor element **101** is a layered body having a plurality of solid electrolyte layers (layers **1** to **6**), but is not limited thereto. The element body of the sensor element **101** may include at least one oxygen-ion-conductive solid electrolyte layer. For example, in FIG. **1**, the layers **1** to **5** other than the second solid electrolyte layer **6** may be layers (e.g., layers composed of alumina) composed of a material other than that of solid electrolyte layers. In this case, the electrodes of the sensor element **101** may be disposed in the second solid electrolyte layer **6**. For example, the measurement electrode **44** in FIG. **1** may be

disposed on the lower surface of the second solid electrolyte layer **6**. Also, the reference gas inlet space **43** may be provided in the spacer layer **5** instead of the first solid electrolyte layer **4**, the reference gas inlet layer **48** may be provided between the second solid electrolyte layer **6** and the spacer layer **5** instead of between the first solid electrolyte layer **4** and the third substrate layer **3**, and the reference electrode **42** may be provided rearward of the third internal cavity **61** and on the lower surface of the second solid electrolyte layer **6**.

[0134] In the above-described embodiment, the controller **96** sets (feedback-controls) the target value $V0^*$ of the voltage $V0$ based on the pump current I_{p1} so that the pump current I_{p1} reaches the target value I_{p1}^* , and feedback-controls the voltage V_{p0} so that the voltage $V0$ reaches the target value $V0^*$, but may perform another control. For example, the controller **96** may feedback-control the voltage V_{p0} based on the pump current I_{p1} so that the pump current I_{p1} reaches the target value I_{p1}^* . In other words, the controller **96** may omit acquisition of the voltage $V0$ from the main pump control oxygen partial pressure detection sensor cell **80** and setting of the target value $V0^*$, and may directly control the voltage V_{p0} (eventually, control the pump current I_{p0}) based on the pump current I_{p1} .

[0135] In the above-described embodiments, the sensor element **101** may include a porous protective layer that covers at least part (e.g., the distal end of the element body) of the element body. The porous protective layer may contain, as constituent particles, ceramic particles of at least one of alumina, zirconia, spinel, cordierite, titania and magnesia. The porous protective layer may cover the gas inlet **10** and the outer pump electrode **23**. When the porous protective layer covers the gas inlet **10**, the aforementioned diffusion resistances A , B also include the diffusion resistance of the porous protective layer. Also, in this case, the aforementioned limiting current values L_a , L_b have values correlated with the diffusion resistances A , B .

Claims

1. A gas sensor comprising a sensor element, and a control device, the sensor element including: an element body that includes an oxygen-ion-conductive solid electrolyte layer, and is internally provided with a measurement-object gas flow portion that introduces a measurement-object gas and causes the measurement-object gas to flow therethrough; a measurement pump cell including an inner measurement electrode disposed in a measurement chamber of the measurement-object gas flow portion, the measurement pump cell being configured to pump oxygen from the measurement chamber to an outside of the element body; an adjustment pump cell including an inner adjustment electrode disposed in an oxygen concentration adjustment chamber located upstream of the measurement chamber in the measurement-object gas flow portion, the adjustment pump cell being configured to adjust an oxygen concentration in the oxygen concentration adjustment chamber; and a reference electrode disposed inside the element body so as to be in contact with a reference gas which serves as a reference for detection of a specific gas concentration that is a concentration of a specific gas in the measurement-object gas, wherein the control device is configured to perform an adjustment-pump control process and a measurement-pump control process, the adjustment-pump control process for adjusting an oxygen concentration in the oxygen concentration adjustment chamber by controlling the adjustment pump cell so that an adjustment voltage that is a voltage across the reference electrode and the inner adjustment electrode reaches an adjustment voltage target value, the measurement-pump control process for pumping out oxygen from the measurement chamber by controlling the measurement pump cell so that a measurement voltage that is a voltage across the reference electrode and the inner measurement electrode reaches a measurement voltage target value, and let $V1^*$ [mV] be the adjustment voltage target value, then following Expression (1) and Expression (2) are satisfied: $V1^* \leq -25.7 * R + 436$. (1)

$V1^* \geq -35.1 * R + 406.6$ (2) where $R = L_b * L_b / L_a$, L_a is a limiting current value [mA] of an

adjustment pump current which flows through the adjustment pump cell when the adjustment pump cell pumps out oxygen from the oxygen concentration adjustment chamber with the sensor element exposed to the measurement-object gas which is a nitrogen-based gas with an oxygen concentration of 20.5%, a water concentration of 3%, and an NO concentration of 0 ppm, and L_b is a limiting current value [μA] of a measurement pump current which flows through the measurement pump cell when the measurement pump cell pumps out oxygen from the measurement chamber with the sensor element exposed to the measurement-object gas which is a nitrogen-based gas with an oxygen concentration of 0%, a water concentration of 3%, and an NO concentration of 500 ppm.

2. The gas sensor according to claim 1, wherein following Expression (3) and Expression (4) are satisfied: $V1 \leq -18.9 * R + 419.2$ (3) $V1 \geq -16.2 * R + 403.4$. (4)

3. The gas sensor according to claim 1, wherein the oxygen concentration adjustment chamber includes a first internal cavity, and a second internal cavity provided downstream of the first internal cavity and upstream of the measurement chamber, the adjustment pump cell includes a main pump cell configured to adjust an oxygen concentration in the first internal cavity, and an auxiliary pump cell configured to adjust an oxygen concentration in the second internal cavity, the inner adjustment electrode includes an inner main pump electrode that is disposed in the first internal cavity and constitutes part of the main pump cell, and an inner auxiliary pump electrode that is disposed in the second internal cavity and constitutes part of the auxiliary pump cell, the adjustment voltage is a voltage across the reference electrode and the inner auxiliary pump electrode, and the limiting current value L_a is a limiting current value of a main pump current which flows through the main pump cell when the main pump cell pumps out oxygen from the first internal cavity with the sensor element exposed to the measurement-object gas which is a nitrogen-based gas with an oxygen concentration of 20.5%, a water concentration of 3%, and an NO concentration of 0 ppm.

4. A method for determining an adjustment voltage target value for control of a sensor element, the sensor element including: an element body that includes an oxygen-ion-conductive solid electrolyte layer, and is internally provided with a measurement-object gas flow portion that introduces a measurement-object gas and causes the measurement-object gas to flow therethrough; a measurement pump cell including an inner measurement electrode disposed in a measurement chamber of the measurement-object gas flow portion, the measurement pump cell being configured to pump oxygen from the measurement chamber to an outside of the element body; an adjustment pump cell including an inner adjustment electrode disposed in an oxygen concentration adjustment chamber located upstream of the measurement chamber in the measurement-object gas flow portion, the adjustment pump cell being configured to adjust an oxygen concentration in the oxygen concentration adjustment chamber; and a reference electrode disposed inside the element body so as to be in contact with a reference gas which serves as a reference for detection of a specific gas concentration that is a concentration of a specific gas in the measurement-object gas, wherein the adjustment voltage target value is a target value of an adjustment voltage when a specific gas concentration that is a concentration of a specific gas in the measurement-object gas is detected using the sensor element, the adjustment voltage being a voltage across the reference electrode and the inner adjustment electrode at a time of feedback control of the adjustment pump cell to adjust the oxygen concentration in the oxygen concentration adjustment chamber, the method including a step (a) of determining the adjustment voltage target value so that lower a diffusion resistance B relative to a diffusion resistance A, the adjustment voltage target value tends to be lower, where the diffusion resistance A is a diffusion resistance from an outside of the sensor element to the oxygen concentration adjustment chamber, and the diffusion resistance B is a diffusion resistance from the outside of the sensor element to the measurement chamber.

5. The method for determining an adjustment voltage target value for control of a sensor element according to claim 4, wherein the step (a) includes: a step (a1) of measuring, as a value correlated

with the diffusion resistance A, a limiting current value I_a of an adjustment pump current which flows through the adjustment pump cell when the adjustment pump cell pumps out oxygen from the oxygen concentration adjustment chamber, and as a value correlated with the diffusion resistance B, a limiting current value I_b of a measurement pump current which flows through the measurement pump cell when the measurement pump cell pumps out oxygen from the measurement chamber; and a step (a2) of determining the adjustment voltage target value so that for the limiting current values I_a , I_b measured in the step (a1), greater the limiting current value I_b relative to the limiting current value I_a , the adjustment voltage target value tends to be lower.

6. The method for determining an adjustment voltage target value for control of a sensor element according to claim 5, wherein in the step (a2), the adjustment voltage target value is determined so that the adjustment voltage target value tends to be lower for a greater value R expressed below:

$$R = I_b \cdot I_b / I_a.$$

7. The method for determining an adjustment voltage target value for control of a sensor element according to claim 5, wherein in the step (a2), the adjustment voltage target value is determined as a value corresponding to the limiting current values I_a , I_b measured in the step (a1) based on a prepared corresponding relationship in advance between the limiting current values I_a , I_b and the adjustment voltage target value.

8. The method for determining an adjustment voltage target value for control of a sensor element according to claim 5, wherein the oxygen concentration adjustment chamber includes a first internal cavity, and a second internal cavity provided downstream of the first internal cavity and upstream of the measurement chamber, the adjustment pump cell includes a main pump cell configured to adjust an oxygen concentration in the first internal cavity, and an auxiliary pump cell configured to adjust an oxygen concentration in the second internal cavity, the inner adjustment electrode includes an inner main pump electrode that is disposed in the first internal cavity and constitutes part of the main pump cell, and an inner auxiliary pump electrode that is disposed in the second internal cavity and constitutes part of the auxiliary pump cell, the adjustment voltage is a voltage across the reference electrode and the inner auxiliary pump electrode, and the limiting current value I_a is a limiting current value of a main pump current which flows through the main pump cell when the main pump cell pumps out oxygen from the first internal cavity.

9. A method for determining an adjustment voltage target value for control of a sensor element, the sensor element including: an element body that includes an oxygen-ion-conductive solid electrolyte layer, and is internally provided with a measurement-object gas flow portion that introduces a measurement-object gas and causes the measurement-object gas to flow therethrough; a measurement pump cell including an inner measurement electrode disposed in a measurement chamber of the measurement-object gas flow portion, the measurement pump cell being configured to pump oxygen from the measurement chamber to an outside of the element body; an adjustment pump cell including an inner adjustment electrode disposed in an oxygen concentration adjustment chamber located upstream of the measurement chamber in the measurement-object gas flow portion, the adjustment pump cell being configured to adjust an oxygen concentration in the oxygen concentration adjustment chamber; and a reference electrode disposed inside the element body so as to be in contact with a reference gas which serves as a reference for detection of a specific gas concentration that is a concentration of a specific gas in the measurement-object gas, wherein the sensor element is configured to, when a specific gas concentration that is a concentration of a specific gas in the measurement-object gas is detected, perform an adjustment-pump control process and a measurement-pump control process, the adjustment-pump control process for adjusting the oxygen concentration in the oxygen concentration adjustment chamber by controlling the adjustment pump cell so that an adjustment voltage that is a voltage across the reference electrode and the inner adjustment electrode reaches the adjustment voltage target value, the measurement-pump control process for pumping out oxygen from the measurement chamber by controlling the measurement pump cell so that a measurement voltage that is a voltage across the

reference electrode and the inner measurement electrode reaches a measurement voltage target value, the method including: a step (b1) of setting the measurement voltage target value to a predetermined value, setting the adjustment voltage target value to a temporary target value, and measuring a first measurement pump current and a second measurement pump current, the first measurement pump current being caused to flow through the measurement pump cell when the adjustment-pump control process and the measurement-pump control process are performed on a first gas as the measurement-object gas, the first gas having a water concentration of a first concentration Ch_1 , the second measurement pump current being caused to flow through the measurement pump cell when the adjustment-pump control process and the measurement-pump control process are performed on a second gas as the measurement-object gas, the second gas having a water concentration of a second concentration Ch_2 lower than the first concentration Ch_1 ; and a step (b2) of changing the temporary target value and performing the step (b1) for multiple times, and determining the temporary target value to be the adjustment voltage target value for control of the sensor element, the temporary target value causing a discrepancy between the first measurement pump current and the second measurement pump current to fall within a predetermined minute range.

10. A method for manufacturing a sensor element, including the method for determining an adjustment voltage target value for control of a sensor element according to claim 4, and being associated with the adjustment voltage target value, the method comprising: a manufacturing process for manufacturing the sensor element; a target value determination process for determining the adjustment voltage target value by performing, on the manufactured sensor element, the steps of the method for determining the adjustment voltage target value for control of the sensor element; and an association process for associating the manufactured sensor element with the determined adjustment voltage target value.

11. A method for manufacturing a gas sensor, comprising: the processes of the method for manufacturing a sensor element according to claim 10; and a connection process for connecting the sensor element associated with the adjustment voltage target value to a control device that feedback-controls the adjustment pump cell so that a voltage across the reference electrode and the inner adjustment electrode reaches the associated adjustment voltage target value.

12. A method for manufacturing a sensor element, including the method for determining an adjustment voltage target value for control of a sensor element according to claim 9, and being associated with the adjustment voltage target value, the method comprising: a manufacturing process for manufacturing the sensor element; a target value determination process for determining the adjustment voltage target value by performing, on the manufactured sensor element, the steps of the method for determining the adjustment voltage target value for control of the sensor element; and an association process for associating the manufactured sensor element with the determined adjustment voltage target value.

13. A method for manufacturing a gas sensor, comprising: the processes of the method for manufacturing a sensor element according to claim 12; and a connection process for connecting the sensor element associated with the adjustment voltage target value to a control device that feedback-controls the adjustment pump cell so that a voltage across the reference electrode and the inner adjustment electrode reaches the associated adjustment voltage target value.
